

Potential of a newly isolated lytic bacteriophage to control *Pseudomonas coronafaciens* pv. *garcae* in coffee plants: Molecular characterization with *in vitro* and *ex vivo* experiments

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ABSTRACT

Traditionally, control of coffee plant bacterial halo blight (BHB) caused by the phytopathogen *Pseudomonas coronafaciens* pv. *garcae* (Pcg) involves frequent spraying of coffee plantations with non-environmentally friendly and potentially bacterial resistance-promoting copper products or with kasugamycin hydrochloride. In this study we report a leap forward in the quest for a new ecofriendly approach, characterizing (both physicochemically and biologically) and testing both *in vitro* and *ex vivo* a new lytic phage for Pcg. An in-depth molecular (genomic and DNA structural features) characterization of the phage was also undertaken. Phage PcgS01F belongs to the class Caudoviricetes, *Drexelviriidae* family and genus *Guelphvirus*, and presents a siphovirus-like morphotype. Phage PcgS01F showed a latency period of 40 min and a burst size of 46 PFU/host cell, allowing to conclude that it replicates well in Pcg IBSBF-158. At Multiplicity Of Infection (MOI, or the ratio of phage to bacteria) 1000, the performance of phage PcgS01F was much better than at MOI 10, promoting increasing bacterial reductions until the end of the *in vitro* inactivation assays, stabilizing at a significant 82 % bacterial load reduction. Phage PcgS01F infected and killed Pcg cells *ex vivo* in coffee plant leaves artificially contaminated, with a maximum of Pcg inactivation of 7.66 log CFU/mL at MOI 1000 after 36 h of incubation. This study provides evidence that the isolated phage is a promising candidate against the causative agent of BHB in coffee plants.

1. Introduction

Managing bacterial diseases of plants is a challenge, with extensive crop losses to phytopathogens being a reality worldwide [1,2]. While conventional disease control strategies involve extensive use of chemicals such as copper-based products and antibiotics (mainly kasugamycin hydrochloride), toxicity due to long-term exposure to copper together with development of antibiotic- and copper-resistant bacterial strains are important drawbacks [1,3–5].

Coffee plant bacterial halo blight is a debilitating disease whose etiological agent is the phytopathogen *Pseudomonas coronafaciens* pv.

garcae (Pcg, an ice-nucleating bacterium that can increase crop susceptibility to frost damage) [6–18], having been reported for the first time back in 1955 in the region of Garça (State of São Paulo), Brazil. The coffee bacterial halo blight disease is characterized by formation of yellowish halos around primary, dark lesions, rupturing infected leaves in adult coffee plants [14,19] and, during the flowering period, development of necrosis in the rosettes of floral remains that leads to defoliation and drying up of productive branches with failure of fruit development [14,19]. Over the last few years, this disease stood out in Brazilian coffee plantations, seriously affecting the coffee trade [20,21]. Although copper-based formulations and antibiotics have regulatory

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approval for use by official regulatory agencies, they are non-environmentally friendly and potentially bacterial resistance-promoting, hence the search for new effective and environmentally-friendly approaches to control Pcg is vital for establishing sustainable agriculture practices [22–27].

Consumers are increasingly demanding for more sustainable and healthier crop products, produced without (or with less) spraying of agrochemicals [28–31], and therefore the development of safer and eco-friendly alternatives to conventional agrochemical treatments of the bacterial halo blight on coffee tree plantations is highly desirable. The use of bacteriophages, viruses that specifically prey bacterial cells, appear therefore as a tailored and targeted approach, with the added advantages that these entities do not scavenge the beneficial microflora and simply disappear once their target (pathogen) bacterial hosts are eliminated [3,32]. While over the last decade several researchers worldwide have attempted to use bacteriophages to control phytopathogens [3,33–42], to the best of our knowledge only one other research work has reported the isolation and characterization of phages to control the phytopathogen Pcg [33], deploying first *in vitro* and *ex vivo* strategies to use such phages to control that pathogen.

In a previous work [33], two new myoviruses lytic for the same phytopathogen were quite effective at MOI 10 (reduction of 4.59 log CFU/mL by phage ph002F and reduction of 3.85 log CFU/mL by phage ph004F) after 10 h of incubation with the host. As a cocktail, the bacterial reduction *in vitro* promoted by the two myoviruses was even larger (5.26 log CFU/mL at MOI 0.1 or 5.03 log CFU/mL at MOI 10), after a 12h-treatment. In *ex vivo* phage-bacteria inactivation experiments, the maximum of Pcg inactivation was 5.36 log CFU/mL at MOI 100 by the cocktail of both phages after 6 h of incubation, and 5.11 log CFU/mL for MOI 100 after a 24h-treatment [33]. But, with those two myoviruses, a slight bacterial regrowth was observed until the end of the incubation timeframe, which did not happen with the newly isolated siphovirus described in the present work at a higher MOI (*ex vivo* bacterial reduction of 7.66 log CFU/mL at MOI 1000 after 36 h of incubation). Although phage ph002F [33] and the phage described herein (phage PcgS01F) were isolated in the very same location, the former was isolated from coffee plant leaves with symptoms of the disease whereas the later was isolated from soil around a diseased coffee plant with approximately 50 years old.

The two myoviruses described by Silva et al. [33] underwent an in-depth molecular characterization [43] and revealed very different genome sizes (165,282 bp for phage ph002F and 151,205 bp for phage ph004F) compared with that of the siphovirus (phage PcgS01F) described in the present work (44,679 bp).

In the research work entertained herein we follow the work by Silva et al. [33], reporting the biological and molecular characterization of a new lytic phage capable of preying on *Pseudomonas coronafaciens* pv. *garcae* (Pcg), together with *in vitro* and *ex vivo* phage-Pcg inactivation trials, aiming at developing a phage-based biocontrol strategy for the management of coffee plant bacterial halo blight.

2. Materials and methods

2.1. Biological material

The bacterial strain used as phage host, *Pseudomonas coronafaciens* pv. *garcae* (Pcg) IBSBF-158 (previously, *Pseudomonas syringae* pv. *garcae*), was acquired from the Phytobacteria Culture Collection of Instituto Biológico (IBSBF, Campinas/SP, Brazil). In a recent publication, Dutta and colleagues [44] proposed *Pseudomonas coronafaciens* sp. nov. as a new species in genus *Pseudomonas*, which is diverse from *P. syringae*. In their paper, Dutta et al. [44] propose the elevation of *P. syringae* pv. *coronafaciens* to a species level as *P. coronafaciens*. Hence, the previously named *Pseudomonas syringae* pv. *garcae* was renamed in the present work as *Pseudomonas coronafaciens* pv. *garcae*. The phage reported herein (PcgS01F) was isolated from a soil sample collected around a coffee

plant aged more than 50 years old, with symptoms of bacterial halo blight, collected on February 16th, 2023, at Fazenda Santo Antônio da Bela Vista (23°18'41.1"S, 47°24'13.2"W; Itu city; Brazil).

All bacterial cultures were maintained in Tryptic Soy Agar (TSA) at 4 °C. Before each assay, one isolated colony was transferred to 25 mL Tryptic Soy Broth (TSB) and grown overnight at the optimum temperature for bacterial growth. An aliquot of this culture (100 µL) was transferred to 10 mL of fresh TSB and grown overnight at the optimum temperature for bacterial growth to reach an optical density (OD_{610 nm}) of 1.0, corresponding to approximately 10⁹ cells per mL.

2.2. Chemicals

Bacteriologic solid agar was purchased from Gibco Diagnostics (Madison WI, USA). The sterilizing filtration systems Sartolab®-RF 250 (with 0.22 µm pore diameter polyethersulphate (PES) membrane) used throughout the work were purchased from Sartorius Biohit Liquid Handling Oy (Helsinki, Finland). Tap water was ultrapurified to a final resistivity of 18.18 MΩ cm and conductivity of 0.05 µS cm⁻¹ in a Master System All MS2000 (Gehaka, São Paulo/SP, Brazil). TSA and TSB culture media, PEG 8000, uranyl acetate, SDS, EDTA, phenol, chloroform, NaOAc, absolute ethanol, K₂HPO₄, KH₂PO₄, MgSO₄, CaCl₂ and NaCl were purchased from Sigma-Aldrich (St. Louis MO, USA). DNase-I, RNase and proteinase K were purchased from Transgen Biotech (Beijing, China). Nuclease-free ultrapure water was purchased from ThermoScientific (Waltham, MA, USA).

2.3. Preparation of a Pcg IBSBF-158 growth curve

The host bacterium was plated on TSA and incubated at 28 °C for 12 h. A single CFU was then withdrawn with a sterile loop (ca. 10 µL inoculum), inoculated in 250 mL sterile TSB and incubated at 28 °C for 24 h. At pre-determined time intervals, the optical density of the culture was evaluated spectrophotometrically at 610 nm. Three independent experiments were performed (each using a 250 mL-Erlenmeyer), and for each experiment the optical density was measured in triplicate for each sampling time.

2.4. Phage enrichment, isolation and enumeration

The bacteriophage was isolated from diseased coffee plant leaves using the enrichment procedure, according to Silva et al. [33], Balcão et al. [45,46], Harada et al. [47] and Kharina et al. [48], with modifications. Coffee plant leaves were grinded and macerated, the resulting leaf tissue was transferred into a 250-mL Schott flask and added with 50 mL SM-buffer (40.19 mM K₂HPO₄, 22.05 mM KH₂PO₄, 85.55 mM NaCl, 1.00 mM MgSO₄ and 1.00 mM CaCl₂), after which it was incubated at room temperature for 2 h, without shaking. After this time period, 50 µL of an overnight culture of Pcg IBSBF-158 (in exponential growth, with OD_{610 nm} ≈ 0.5) were added to the Schott flask and supplemented with 50 mL TSB. Following incubation at 28 °C for 24–48 h, the enriched sample was centrifuged (9000 rpm, 4 °C, 10 min) and the supernatant was carefully collected and mixed with chloroform (1:1, v/v) to remove any bacterial cells present. The upper aqueous phase was then filtered through sterile 0.22 µm PES membrane, and tested via spot-testing 10 µL-droplets of the filtered supernatant onto a Pcg IBSBF-158 lawn [33,45,46]. The Petri plate was then incubated at 28 °C during 24 h. Following incubation, the plate was checked for the presence of either clear or turbid lysis zones, indicative of the presence of phages.

Isolation of a phage plaque was carried out following the procedure described by Silva et al. [33], Balcão et al. [45,46] and Harada et al. [47], with modifications. For isolation of a single phage plaque, the procedure was sequentially repeated three times. The titre of the resulting phage suspension (PFU/mL, plaque forming units/mL) was then determined [33,45–47] by plating serial dilutions of the phage

suspension via the double-layer agar method.

2.5. PEG-precipitation of phage virions

A sterile mixture of PEG 8000 (10 %, w/v) and NaCl (1 M), in a volumetric proportion of 2:1, respectively, was added to the phage suspension (10^{14} PFU/mL). The resulting mixture was incubated overnight at 4 °C, centrifuged at 11,000 rpm (4 °C, 45 min) and the supernatant discarded, with the pellet being resuspended and homogenized in 5 mM MgSO₄.

2.6. Transmission electron microscopy (TEM) analyses

Phage particles of PEG-concentrated suspension were firstly centrifuged (4 °C, 150 min, 45,000 rpm, $124,740 \times g$) in a benchtop Beckman-Coulter ultracentrifuge (model Optima TLX micro-ultracentrifuge) equipped with a TLA-55 Fixed-Angle Rotor (Indianapolis IN, USA). Negative staining of the phage virions with uranyl acetate at 2 % (w/v) [47] allowed observation of their structure. The TEM analyses were performed in a Transmission Electron Microscope from JEOL (model JEM 2100, Tokyo, Japan), equipped with a LaB₆ filament, operating at 200 kV and with resolution of 0.23 nm; a high-resolution (11 MP, 4.0×2.7 k pixels / $9 \times 9 \mu\text{m}^2$) CCD camera from GATAN Inc. (model ORIUS™ 832.J4850 SC1000B, Pleasanton CA, USA) was utilized for the acquisition of digital images, via software Gatan Microscopy Suite (DigitalMicrograph from Gatan Inc., version 2.11.1404.0). Average virion capsid and tail dimensions were determined using the public domain ImageJ software (version 1.52a) from the National Institute of Health (NIH; Bethesda MD, USA), measuring 5 phage particles.

2.7. UV-Vis spectral scans and phage virion extinction coefficient

Determination of the phage virions extinction coefficient followed the procedure described elsewhere [33,45–47,49].

2.8. Phage host range and efficiency of plating (EOP)

Phage host-range was determined via spot testing 5 µL of the phage suspension in lawns of the bacterial strains listed in Table 1, according to the procedure described by Silva et al. [33], Balcão et al. [45,46], Harada et al. [47] and Pinheiro et al. [35,36]. Sensitivity of the target bacteria to the newly isolated phage was inferred from formation of a clear lysis zone at the spot. The bacteria tested were differentiated according to the translucency of the zone at the spot as either + (clear lysis zone) or – (no-lysis zone), and the EOP was determined as $EOP = \frac{\text{Average PFU}_{\text{target bacteria}}}{\text{Average PFU}_{\text{host bacteria}}} \times 100$ [33,35,47,50] for those bacteria with positive spot tests (isolation host, EOP=100 %).

2.9. Phage adsorption rate onto host cells

The adsorption parameters for phage PcgS01F were obtained from the virion adsorption curve using Pcg IBSBF-158 and PcgS01F at $MOI \leq 0.001$ (to ensure that all phage particles would be adsorbed to bacterial cells [51]) as described by Silva et al. [33], Balcão et al. [45,46] and Harada et al. [47], in three independent experiments.

Assuming that the phage virions in suspension can detect, and adsorb onto, receptor moieties decorating the surface of susceptible bacterial cells, producing a reversible phage virion-bacteria complex that may, or not, infect the bacterium, the mechanistic rationale postulated by Watanabe et al. [52], Moldovan et al. [53] and Storms and Sauvageau [54], with modifications [33,55], leads to the mathematical model $P_t/P_0 = (\phi/(\delta \times X_0 + \phi)) \times \{1 + (\delta \times X_0/(\phi)) \times e^{-(\delta \times X_0 + \phi) \times t}\}$. This model was used to estimate the virion adsorption and desorption rate constants, via nonlinear fitting to the experimental phage virion

Table 1

Host range of phage PcgS01F evaluated on 28 bacterial strains with indication of clear lysis zone (+) and no-lysis zone (–) produced in the spot-test. For the host strain, EOP was considered as 100 %.

Bacterial strains	Source	Phage PcgS01F			
		Spot test	PFUs in isolation host	PFUs in target bacteria	EOP (%)
<i>A. baumannii</i> ATCC 19606	Collection, ATCC	–			
<i>A. hydrophyla</i> ATCC 7966	Collection, ATCC	–			
<i>B. subtilis</i> CCCD-B010	Collection, CEFAR	–			
<i>C. freundii</i> CCCD-C001	Collection, CEFAR	–			
<i>E. aerogenes</i> CCCD-E001	Collection, CEFAR	–			
<i>E. faecalis</i> CCCD-E002	Collection, CEFAR	–			
<i>E. coli</i> ATCC 25922	Collection, ATCC	+		5.92×10^{13}	10.73
<i>K. pneumoniae</i> CCCD-K001	Collection, CEFAR	–			
<i>K. pneumoniae</i> NCTC-13439	National Collection of Type Cultures, UK Health Security	–			
<i>P. mirabilis</i> CCCD-P001	Collection, CEFAR	–			
<i>P. vulgaris</i> CCCD-P002	Collection, CEFAR	–			
<i>P. aeruginosa</i> CCCD-P004	Collection, CEFAR	–			
<i>Pseudomonas syringae</i> pv. <i>actinidiae</i> (Psa) CRA-FRU 14.10	Collection, CRA-FRU	+		3.38×10^{10}	0.0061
Pcg IBSBF-152	Collection, IBSBF (Campinas/SP, Brazil)	–			
Pcg IBSBF-1197	Collection, IBSBF (Campinas/SP, Brazil)	–			
Pcg IBSBF-1293	Collection, IBSBF (Campinas/SP, Brazil)	–			
Pcg IBSBF-1372	Collection, IBSBF (Campinas/SP, Brazil)	–			
Pcg IBSBF-158	Collection, IBSBF (Campinas/SP, Brazil)	+	5.52×10^{14}		100 (host)
Pcg IBSBF-2212	Collection, IBSBF (Campinas/SP, Brazil)	–			
Pcg IBSBF-248	Collection, IBSBF (Campinas/SP, Brazil)	–			
Pcg IBSBF-2511	Collection, IBSBF (Campinas/SP, Brazil)	–			
Pcg IBSBF-2840	Collection, IBSBF (Campinas/SP, Brazil)	–			

(continued on next page)

Table 1 (continued)

Bacterial strains	Source	Phage PcgS01F			EOP (%)
		Spot test	PFUs in isolation host	PFUs in target bacteria	
Pcg IBSBF-3024	Collection, IBSBF (Campinas/ SP, Brazil)	-	—	—	—
Pcg IBSBF-3046	Collection, IBSBF (Campinas/ SP, Brazil)	-	—	—	—
Pcg IBSBF-3270	Collection, IBSBF (Campinas/ SP, Brazil)	-	—	—	—
<i>S. enteritidis</i> CCD- S004	Collection, CEFAR	-	—	—	—
<i>S. typhimurium</i> ATCC 13311	Collection, ATCC	-	—	—	—
<i>X. citri</i> pv. <i>citri</i> IAPAR 306	Collection, IAPAR (Paraná/ PR, Brazil)	-	—	—	—

adsorption data set, where P_t/P_0 is the normalized phage concentration (dimensionless), δ is the (first order) phage adsorption rate onto susceptible bacterial host cells ($\text{CFU}^{-1} \text{ mL min}^{-1}$), ϕ is the (first order) phage desorption rate from reversible phage-bacteria complexes (min^{-1}), X_0 is the initial concentration of susceptible (uninfected) bacterial cells (CFU mL^{-1}) and t is the infection time (min).

2.10. Phage virion one-step growth curve

The growth parameters for phage PcgS01F were determined from the growth curve in a single synchronous cycle using Pcg IBSBF-158 and PcgS01F at $\text{MOI} \leq 0.001$ as described by Silva et al. [33], Balcão et al. [45,46] and Harada et al. [47], in three independent experiments. A typical sigmoidal model (4-parameter logistic regression), viz. $\text{Log}(P_t) = P_\infty + \left\{ (P_0 - P_\infty) / \left(1 + \left(\frac{t}{\psi} \right)^\xi \right) \right\}$, was adjusted via nonlinear fitting to the experimental phage growth data (PFU/mL) and allowed estimation of the phage growth parameters (latent, eclipse and intracellular accumulation periods, and virion morphogenesis yield). In this model, P_t , P_0 and P_∞ are the phage concentrations (PFU/mL) at times t , 0 and ∞ , respectively, ψ is the inflection point of the curve, ξ is the steepness of the curve (Hill's slope) and t is the time (min).

2.11. In-vitro phage-bacteria inactivation experiments

Inactivation of planktonic Pcg IBSBF-158 cells (since bacteria alternate between two modes of growth, a unicellular life phase in which the cells are freely-swimming (planktonic), and a multicellular life phase in which the cells are sessile and live in a biofilm [56], and because all *in vitro* phage-bacteria inactivation assays involved bacterial suspensions, the term planktonic is used here to designate the free cells in suspension) (concentration adjusted to 10^5 CFU/mL , in exponential growth) by phage PcgS01F was studied at MOI values 0.01, 0.1, 1, 10, 100 and 1000. For each MOI experiment (conducted in three independent assays), a bacterial control (BC) was also prepared, encompassing only planktonic host cells. Both BC and treatments (BP-B, bacteria and phage – bacterial concentration) were incubated with the same time/temperature parameters, viz. 28°C and 12 h. Two mL-aliquots of BC and treatment samples (BP-B) were withdrawn at predetermined intervals of time up to a total treatment timeframe of 12 h, and their absorbance duly measured at a wavelength of 610 nm.

2.12. Assessment of the frequency of emergence of bacterial mutant cells resistant to the phage

Development of resistant mutants of Pcg IBSBF-158 cells to phage PcgS01F was studied following the procedure described by Silva et al. [33], Pinheiro et al. [35] and Filippov et al. [57]. The frequency of Pcg IBSBF-158 spontaneous phage-resistant mutant cells was calculated dividing the number of bacteria that emerge in the presence of phage virions (i.e., the resistant bacteria) by the total number of sensitive bacteria (from a culture without phages) [57]. Three independent assays were carried out.

2.13. Study of the impact of abiotic factors upon phage lytic viability

The impact of T, pH, and solar radiation upon phage PcgS01F ($10^{4-6.5} \text{ PFU/mL}$) lytic viability was studied in phosphate-buffered saline (PBS) (10 mL). For the pH and T experiments, aliquots were withdrawn during a period of up to 72 h of incubation. For the solar radiation experiments, aliquots were withdrawn during a period of up to 7 h of exposure to direct sunlight. For each condition, phage virion concentrations were determined in triplicate in three independent experiments.

2.13.1. pH

The impact of pH upon phage PcgS01F lytic viability was studied at constant temperature (25°C) by adding concentrated phage suspensions to sterile PBS at different pH values (viz. 5.0, 7.0 and 10.0).

2.13.2. Temperature

The impact of T upon phage PcgS01F lytic viability was studied by adding concentrated phage suspensions to sterile PBS (pH 7.0) followed by incubation at a constant temperature (either 4, 17, 25 or 40°C).

2.13.3. Solar radiation

The impact of solar radiation upon phage PcgS01F lytic viability was studied by adding concentrated phage suspensions to sterile PBS (pH 7.0) followed by outside exposure to natural sunlight (these were the test samples, SR). Control samples (SR-C) were prepared in exactly the same way but were not exposed to solar radiation, being left outside at the same temperature as test samples. These studies were carried out on a shiny day (direct solar irradiance of ca. 1636.4 kWh/m^2 (data obtained from <https://globalsolaratlas.info/map?c=-23.500416,-47.399797,18&s=-23.50034,-47.400786&m=site>)) with ambient T varying from 29°C (09h00) to 35°C (end of the experiment). Because labware made of borosilicate glass is impervious to most UV-radiation wavelengths [55], these experiments were carried out in small colorless flasks made of common (non-mineral) glass oriented towards the sun in such a way that UV-A and UV-B radiation wavelengths from sunlight were allowed to pass through and hit both the surface of the phage suspension and the lateral parts of the glass flasks.

2.14. Whole phage genome extraction, purification and sequencing

2.14.1. Whole phage genome extraction and purification

Five hundred microlitres of PEG-concentrated phage suspension was treated with 1.25 μL DNase-I (20 mg/mL) and 1.25 μL RNase (10 mg/mL) at 37°C for 1 h. After incubation, 1.25 μL proteinase K (20 mg/mL), 25 μL aqueous SDS (10 %, w/w) (final SDS concentration of 0.5 %, w/w) and 20 μL aqueous EDTA (0.5 M, pH 8.0) (final EDTA concentration of 20 mM) were added to the sample, further incubated at 60°C for 1 h, and allowed to cool down to room temperature. DNA extraction was performed using the phenol: chloroform protocol as briefly described. Phenol was added to the mixture in a 1:1 (v/v) proportion, centrifuged at 6000 rpm for 5 min, and the supernatant added with an equal volume of chloroform; this last step was repeated twice. Following centrifugation (6000 rpm for 5 min), the aqueous phase was carefully collected, and DNA was precipitated via addition to the supernatant of 1/10 of its

volume of 3 M NaOAc (pH 7.5) and 2.5x of its volume of cold absolute ethanol. The mixture thus produced was incubated overnight at -20°C , at -86°C during 30 min, and then centrifuged at 14,000 rpm during 20 min. The supernatant was discarded, and the pellet was allowed to dry, after which it was dissolved in 50 μL of nuclease-free ultrapure water. Purified phage DNA underwent a final clean-up step using RNase and stored at -20°C . DNA purity and concentration were evaluated on a DS-11FX spectrophotometer (DeNovix Inc., Wilmington DE, USA) at 260 nm, 280 nm and 230 nm.

2.14.2. Whole phage genome sequencing

Purified dsDNA of the isolated phage was sequenced at NGS SOL-UÇ ÔES GENÔMICAS (Piracicaba/SP, Brazil) using the Illumina MiSeq platform and 2–20 ng of purified dsDNA to prepare the shotgun genomic library with Illumina Nextera DNA library preparation kit (Illumina, San Diego CA, USA). The DNA fragment library was cleaned up with Agencourt AMPure XP beads (Beckman Coulter, Indianapolis IN, USA) and average fragment size (400–700 bp) was verified by running in the 2100 Bioanalyzer using a Agilent High Sensitivity DNA chip (Agilent, Palo Alto CA, USA). Quantification of Illumina sequencing library via quantitative PCR, clusterization, normalization, and sequencing was performed following standard protocols for sequencing in the Illumina MiSeq platform. The library was subjected to one run using the MiSeq Reagent kit v3 (600-cycle format, paired-end (PE) reads).

2.15. Phage genome assembly, taxonomy, annotation, and phylogeny

2.15.1. Evaluation of sequencing reads

The sequencing data obtained by NextSeq 2000 (Illumina, Inc., San Diego CA, USA) was analyzed using the software BCL Convert version 3.8.4, which base calls the sequencing images and converts them into fastq format sequences, with each base accompanied by a Phred quality score [58]. The quality of sequencing was analyzed using the software FastQC version 0.12.1 (<https://www.bioinformatics.babraham.ac.uk/projects/fastqc/>), with a unified report being obtained with the software MultiQC version 1.17 [59].

2.15.2. Genome assembly

Following processing of the sequencing reads, the phage virion genome was assembled using the software MEGAHIT version 1.2.9 [60, 61], a computationally efficient pipeline specialized in assembling prokaryotic genomes that also uses multiple kmers for the assembly of metagenomes. Standard settings were used for MEGAHIT, together with the parameter “-presets meta-sensitive”, with contigs smaller than 500 bp being discarded. Following assembly, the sequencing reads were mapped onto the assembly contigs using the software minimap2 version 2.25 [62] to estimate the sequencing coverage of each contig.

2.15.3. Taxonomic evaluation

The taxonomic assessment of contigs was carried out using the software Kraken2 version 2.1.2 [63], a taxonomic classifier based on kmers, using the PlusPF database version 20230605 (available at <https://benlangmead.github.io/aws-indexes/k2>). The viral contigs were separated according to the genus identified by the softwares Kraken2 version 2.1.2 [63] and KrakenTools version 1.2 [64], followed by using the software Ccfind version 1.4.5 [65] to identify circular contigs and remove terminal repeats from circular contigs. Finally, the software BLAST+ version 2.14.1 [66] was used to determine the bacteriophage most similar to the retrieved viral contigs.

2.15.4. Genome annotation

The Pharokka annotation pipeline version 1.6.0 [67] and Pharokka Database v1.4.0 (<https://zenodo.org/records/8267900>) (accessed on 04 Jan 2024)) were used to annotate the phage genome. Pharokka is a specialized pipeline for annotating phage genomes, that combines several bioinformatic tools to make the prediction and result in a

functional annotation of several categories of genes: (i) transfer RNAs (tRNAs) and transfer-messenger RNAs (tmRNAs), using the softwares tRNAscanSE version 2.0.12 [68] and ARAGORN version 1.2.41 [69]; (ii) clusters of CRISPRs, using the software MinCED version 0.4.2, a reimplementation of CRT [70]; (iii) protein-coding genes (CDS) predicted with the software PHANOTATE version 1.5.1 [71] and annotated with similarity searches using the software Mmseqs2 version 13.45111 [72] against custom versions of the databases PHROG [73], CARD [74] and VFDB [75]. Pharokka also identifies the phage of the databank INPHARED [76] with the most similar genome relative to the genome of the assembled phage, using the software Mash version 2.3.0 [77] for similarity searches between genomes. At last, if no similar phage genome is identified in the INPHARED databank, the software BLAST+ version 2.14.1 [66] is used to determine the bacteriophage most similar to the recovered viral contig(s), using the NCBI NT database [78]. A circular map of the annotated phage genome was generated using Pharokka (version 1.6.0) [67].

2.15.5. Viral proteomic tree

Using computational time of the SuperComputer System of the Institute for Chemical Research of the Kyoto University (Kyoto, Japan), the web-based ViPTree application (<https://www.genome.jp/viptree> (accessed on 13 Jan 2024)) [79] was used to analyze similarities and relationships between phage PcgS01F and other prokaryotic dsDNA viruses.

2.16. Mechanical features of the PcgS01F phage genome

2.16.1. DNA structural features

The DNA structural features such as Propeller Twist (ProT), Helical Twist (HelT), Roll and the minor groove width (MGW) were computed for the PcgS01F phage genome. These values were obtained using a high-throughput prediction methodology called DNASHape (<http://rohslab.cmb.usc.edu/DNASHape/>) [80], as described elsewhere [43,47,81]. Briefly, each one of the structural features were predicted using a pentamer model at a single-nucleotide position using the sequence context of the next base pair. A Python (version 3.9.12) custom script for plotting the resulting heatmaps data was then created and run in Jupyter Notebook (version 6.4.8) within Anaconda Navigator (version 2.1.4, Anaconda Inc.). Once the predicted values were obtained, data was fully analyzed and its characteristics were obtained for a better understanding of the predicted values.

2.16.2. Correlation of the DNA structural features

After determining the DNA shape features for the phage genome, a Pearson correlation analysis was conducted among the features within the phage genome. This statistical approach aimed to identify any correlation between DNA structural features of the phage genome. The outcomes were visualized through heatmaps plotting the number of nucleotides per genome and the 4 structural features, illustrating the Pearson correlation values, the distribution of values for each feature, and the corresponding statistical significance, with colors like red and blue showing a high and low Pearson correlation respectively.

2.16.3. Dinucleotide frequency in the PcgS01F phage genome

A custom Python script was elaborated to investigate the net occurrence of the 16 dinucleotide combinations (AA, AC, AG, AT, CA, CC, CG, CT, GA, GC, GG, GT, TA, TC, TG, and TT) in the genome of phage PcgS01F, as described by Basu et al. [82]. The occurrence of dinucleotides is defined as the number of times a dinucleotide n appears in the genome, divided by the length of the genome, and

multiplied by percentage. These values were calculated for the phage genome and plotted as a bar plot.

2.16.4. Phage genome cyclizability

The looping capacity, also called cyclizability, associated with the phage genome was assessed following the methodology proposed by Basu et al. [82–84] and Zhang et al. [85] and fully described elsewhere [43,47,81]. Intrinsic cyclizability was calculated using a 50-nucleotide interval with a 7-base pair overlap. The resulting data was graphically represented along the phage genome to observe the correlation between the cyclizability and their genomic positions. Additionally, a heatmap and a histogram were generated for a visual representation of the distribution of the cyclizability values. A Python (version 3.9.12) custom script for calculating genome cyclizability was created and run in Jupyter Notebook (version 6.4.8) within Anaconda Navigator (version 2.1.4, Anaconda Inc.). Cyclizability values were only calculated every 7th base pairs, aiming at checking how the bendability changes around some important locations in the phage genome, and to simply average bendability over

the entire phage genome and compare with different phages published in the literature.

2.17. Ex vivo Pcg-phage inactivation in artificially contaminated coffee plant leaves

Apparently healthy coffee plant leaves (i.e., without signs of bacterial halo blight) were kept at 4 °C before assays. All leaves collected were cut into small square pieces (3 cm×3 cm) and duly disinfected via immersion in H₂O₂ at 3 % (v/v) during 15 min followed by UV irradiation on both sides for an extra 15 min. For these experiments, 28 groups of leaf square pieces (4 groups for each of the 7 incubation timeframes) were used. Each group integrated 3 replicates of leaf square pieces (for a total of 28 groups x 3 replicates = 84 leaf square pieces). Each replicate was maintained in a separate Petri plate. Pcg IBSBF-158 (100 µL) was added to 14 of the 28 groups to obtain a final concentration of 10⁸ CFU/mL (1 mL of fresh overnight bacterial suspension at 10⁹ CFU/mL diluted in 9 mL of PBS). To the other 14 groups one added the same volume of plain TSB. All 28 groups were incubated during 1 h at 28 °C. Of the 14 groups of leaf square pieces infected with Pcg IBSBF-158, seven were

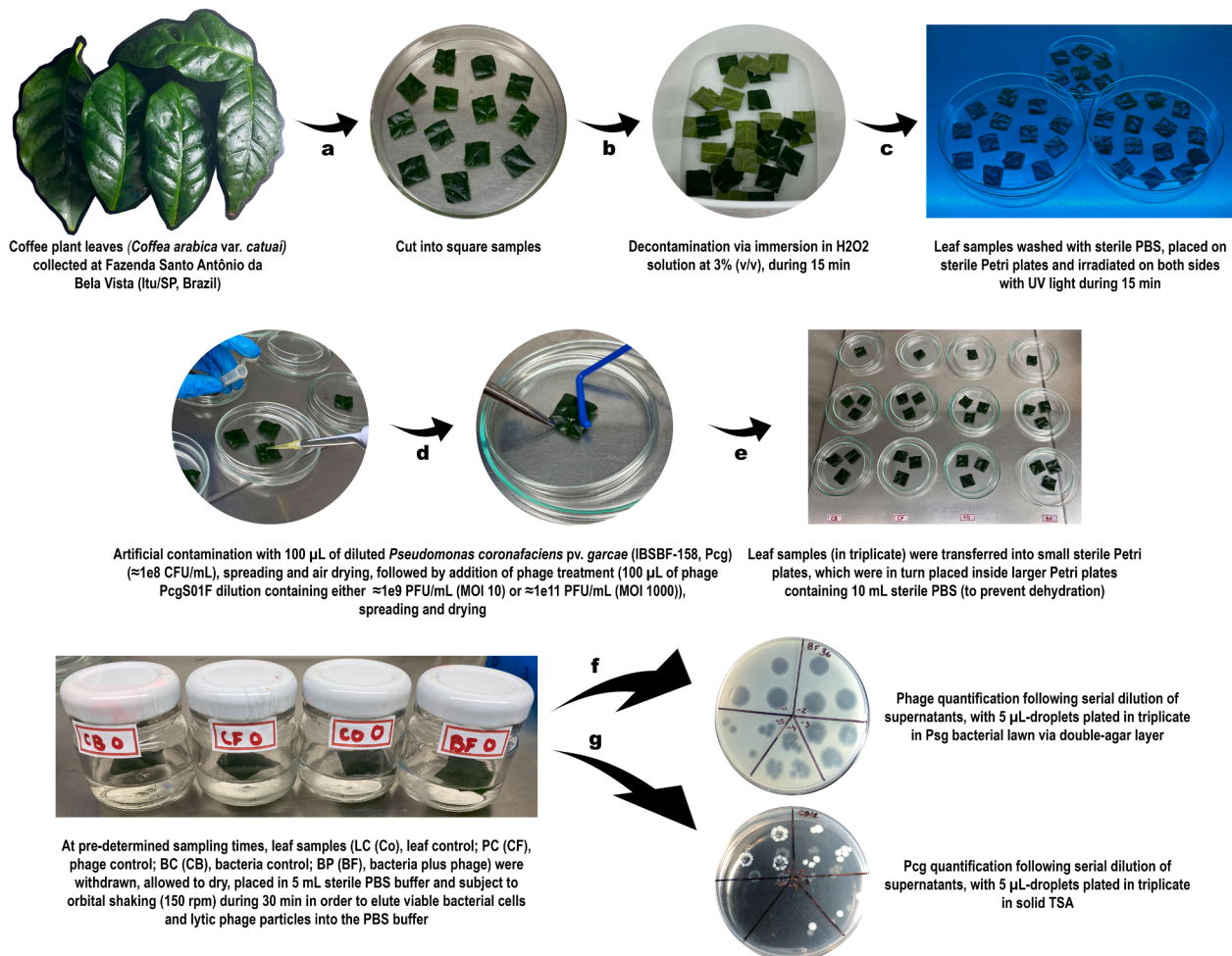


Fig. 1. Simplified scheme of the ex vivo phage treatment assays using phage PcgS01F on artificially contaminated coffee plant leaves. Legend: (a) healthy coffee plant leaves collected from a coffee plantation were cut into small square (3 cm×3 cm) pieces; (b) square pieces of coffee plant leaves were decontaminated using aqueous H₂O₂ at 3 % (v/v); (c) following decontamination, the square pieces of coffee plant leaves were thoroughly washed with sterile PBS buffer and further irradiated on both sides with UV radiation; (d) after being fully decontaminated, the square pieces of coffee plant leaves were artificially contaminated via inoculation with Pcg suspension, allowed to dry, further added with the phage suspension at a given MOI and equally allowed to dry; (e) during the phage treatment, all square pieces of coffee plant leaves were maintained in a moist environment to prevent dehydration and, at pre-determined intervals of time, samples were withdrawn from the moist environment and submerged in PBS buffer under orbital shaking so as to allow transference of viable Pcg cells and phage particles into the supernatant, after which the supernatant was serially diluted for quantification of (f) phage particles via the double agar layer method following incubation at 28 °C during 18 h, and (g) Pcg bacterial concentration via the drop-plate method in TSA medium following incubation at 28 °C during 48 h.

inoculated with phage PcgS01F at a MOI of either 10 or 1000 (these were the treatments, bacteria plus phage, BP) and the remaining seven groups were not inoculated with phage (these were the bacteria control, BC). Regarding the 14 groups of leaf square pieces not infected with Pcg IBSBF-158, seven were inoculated with phage PcgS01F at a MOI of either 10 or 1000 (these were the phage control, PC) and the remaining seven groups were not inoculated with phage PcgS01F (these were the leaf control, LC). In order to keep a moist surface on the leaf square pieces, small Petri plates (60 mm) containing the leaf square pieces were placed inside larger Petri plates (90 mm) containing 10 mL of sterile PBS. All 28 groups of leaf square pieces were incubated exactly under the same conditions. Samples of leaf square pieces of treatment and controls were withdrawn at times 0 and after 3, 6, 9, 12, 24 and 36 h of incubation, immediately soaked in 5 mL of sterile PBS and further incubated at 28 °C (30 min) with orbital shaking (150 rpm). Phage titer was determined in triplicate for all assays via the double agar layer method using 5 μ L-droplet inocula after an incubation period of 18 h at 28 °C, whereas bacterial concentration was determined in triplicate by the drop-plate method in TSA medium using 5 μ L-droplet inocula after an incubation period of 48 h at 28 °C. This assay was repeated three times over the course of three weeks in order to ensure independent replicates. A simplified scheme of the procedure followed is depicted in Fig. 1.

2.18. Statistical analyses

The goodness of the non-linear fittings of both the adsorption and 4-PL mathematical models (i.e., the expectation functions) to the experimental data were statistically assessed via tests of lack of fit of the expectation functions with respect to the phage virion adsorption data onto, and growth data within, Pcg IBSBF-158 cells, following the statistical procedures described by Balcão et al. [45,46,86,87] and Pereira et al. [55].

The normal distribution of the data was checked via Kolmogorov-Smirnov test, whereas the homoscedasticity was assessed via Levene's test. The significance of bacterial concentrations during treatment with the newly isolated lytic phage, in *in vitro* and *ex vivo* phage-bacteria inactivation assays, along the sampling times, was tested via a statistical two-way ANOVA. The significance of the differences recorded for bacterial concentration was evaluated by comparing the results of test samples of each experimental treatment (Pcg IBSBF-158, BP) with the corresponding bacteria control (Pcg IBSBF-158, BC) for the different

incubation times. For different MOI, the significance of the differences was determined by comparing the results obtained in the test samples with the results obtained for the corresponding control samples, for the different inactivation times. A value of $p < 0.05$ was considered to be statistically significant. All statistical analyses were performed using Microsoft Excel (Microsoft, Redmond WA, USA).

3. Results

In this work, one novel isolated lytic phage for *Pseudomonas coronafaciens* pv. *garcae* IBSBF-158 (Pcg IBSBF-158) forming clear plaques of lysis on a lawn of its host, a hallmark feature of virulent phages, was evaluated aiming at its potential future utilization in coffee plantations for the control of Pcg, together with extensive physicochemical, biological and molecular characterization.

3.1. Pcg growth curve

For isolating lytic bacteriophage particles, one used as host a collection bacterium (viz. Pcg IBSBF-158), and therefore obtaining a growth curve (Fig. 2) was mandatory to observe when the cells entered the exponential growth phase. Up to 6 h of growth, approximately, the bacterium remains in a latency period and, after this short timeframe, its exponential growth phase begins and extends up to 14 h of growth (as shown in Fig. 2). Then, the onset of the stationary phase period could be observed. These results were vital to the subsequent research work, since phage infection occurs optimally when the bacterial culture is in the exponential growth phase. Non-linear fitting the Gompertz function to the experimental bacterial growth data allowed to estimate the maximum biomass concentration at $t = \infty$ as 1.105×10^9 CFU/mL (corresponding to a maximum absorbance of 1.1048) and the lag period as 330 min (ca. 5.5 h).

3.2. Phage plaque isolation

Phage PcgS01F was isolated from samples of diseased coffee plant leaves. Its virions formed clear plaques of lysis on the Pcg lawn, with diameters of ca. 1.5 mm (as shown in Fig. 3a). No secondary halo could be observed in the frontier of its lysis plaques (Fig. 3b, zoomed in plaque), indicating that this phage does not produce depolymerase enzymes [88–90]. A high titre phage suspension (10^{14} PFU/mL) was obtained.

Based on the morphological analysis of the phage particles by TEM

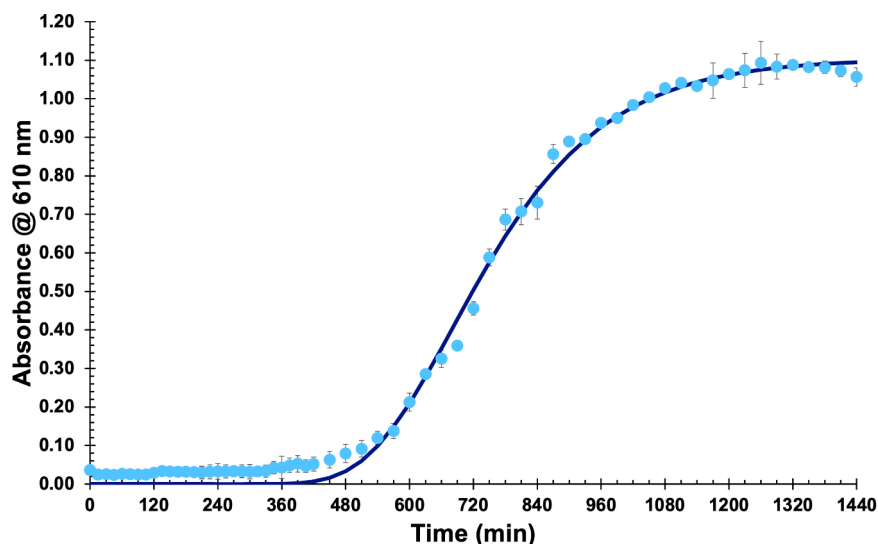


Fig. 2. Growth curve of the host bacteria (Pcg IBSBF-158). The solid blue line represents the non-linear fitting performed (Gompertz function) to the experimental data. Error bars represent the standard deviation ($n = 9$).

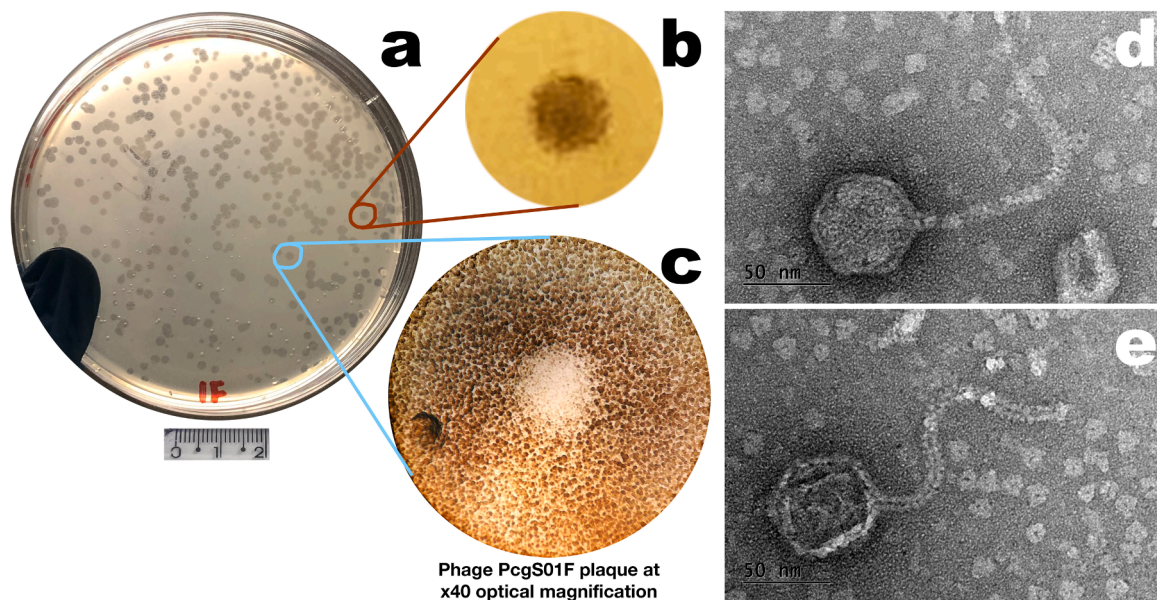


Fig. 3. Morphology of phage PcgS01F (a) plaques on its bacterial host (Pcg IBSBF-158). A lysis plaque is zoomed in (b) to observe the absence of secondary halos. A lysis plaque was also observed under optical microscopy (x40 objective magnification) (c), allowing to depict bacterial debris around the phage plaque. The phage virion morphotype was also obtained by TEM analysis following negative-staining (d,e). The negative-staining TEM photomicrographs of the PcgS01F phage virions allowed to observe the intact head (containing the dsDNA) and bended tail (d), and empty head and bended tail following translocation of its dsDNA (e).

(Fig. 3d,e), phage PcgS01F belongs to the class Caudoviricetes and displays a siphovirus-type morphotype. Phage PcgS01F has a perfect icosahedral capsid (Fig. 3d,e) with approximately $(59.9 \text{ nm} \pm 2.9 \text{ nm}) \times (56.4 \text{ nm} \pm 2.7 \text{ nm})$ (length $\times$ width) and a long, flexible, non-contractile tail with a length $\times$ thickness of approximately $(165.8 \text{ nm} \pm 13.6 \text{ nm}) \times (9.5 \text{ nm} \pm 1.2 \text{ nm})$.

3.3. UV-Vis spectral scan and determination of the phage virion extinction coefficient

The UV-Vis spectral scan of the PEG-concentrated phage PcgS01F suspension is displayed in Fig. 4a, whereas the relationship between the concentration of phage virions and its corrected absorbance is displayed in Fig. 4b. A maximum absorption was observed around 250 nm. By fitting the Beer-Lambert linear relationship to the phage virion concentration as a function of its corrected absorbance (data not shown), the extinction coefficient of the whole phage virions could be determined as $\epsilon_{\text{PcgS01F}} = 2.260 \times 10^{-9} \text{ (PFU/mL)}^{-1} \text{ cm}^{-1}$ (Fig. 4b). The correction that was made to the absorbance data by subtracting $\text{Abs}_{320 \text{ nm}}$ (low absorbance by phage virion structural protein

chromophores at this wavelength) was meant to eliminate light scattering from whole phage virions and non-virion particulates (higher absorbance at this wavelength) [33,35,45–47,49,55].

3.4. Phage host range and efficiency of plating (EOP)

Spot testing indicated that phage PcgS01F, besides its isolation host (Pcg IBSBF-158), could form completely cleared zones of lysis on 2 of the 28 strains tested (Table 1), infecting *E. coli* ATCC 25922 and *Psa* CRA-FRU 14.10 with an EOP of 10.73 % and 0.0061 %, respectively.

The results obtained show that phage PcgS01F could bind to two of the target bacterial strains tested (Table 1) but did not efficiently infect these bacteria, killing them most likely via an abortive infection or low efficiency “lysis from within”, but was not able to produce a large amount of progeny virions (although productively infecting these two bacterial strains), resulting in the quite low value of EOP in *Psa* CRA-FRU 14.10 (Table 1).

3.5. Phage virion adsorption curve

Phage PcgS01F adsorption experiments showed that ca. 50 % of the

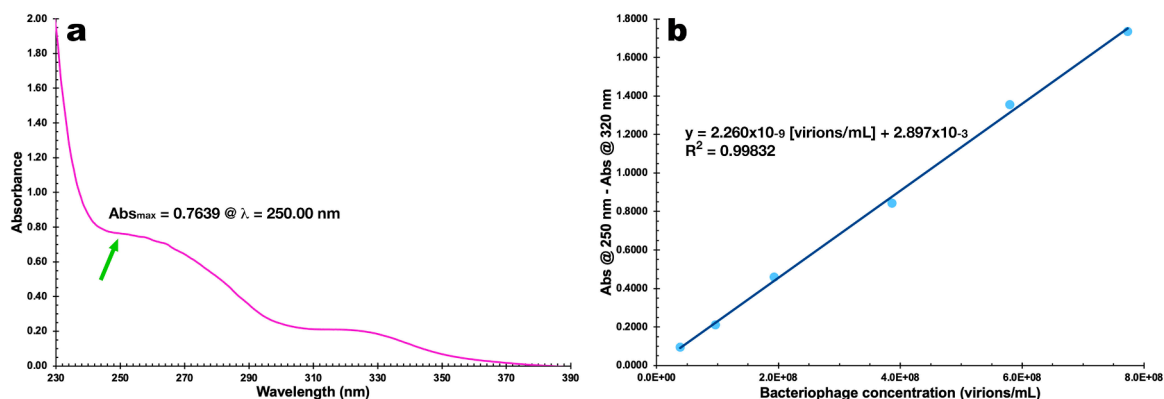


Fig. 4. UV-Vis spectral scan of diluted phage PcgS01F suspension (a) and linear relationship between the concentration of (whole) phage virions and its corrected absorbance (b).

virions adsorb to Pcg cells after 15 min, with more than 90–95 % of them being adsorbed after 40 min (Fig. 5).

Non-linear fitting the normalized phage adsorption decay model to the experimental data (2.10) (r^2 (coefficient of determination) = 0.9886) produced the adsorption rate (δ) of phage PcgS01F virions onto their bacterial host cells ($\delta_{\text{PcgS01F}} = 2.755 \times 10^{-10} \text{ CFU}^{-1} \text{ mL min}^{-1}$) and desorption rate (ϕ) of the phage virions from reversible virion-bacteria complexes ($\phi_{\text{PcgS01F}} = 2.500 \times 10^{-3} \text{ min}^{-1}$).

A statistical analysis of lack of fit of the virion adsorption model [52, 54] to the experimental phage PcgS01F adsorption data allowed to conclude that, at $\alpha = 0.05$ (95 % confidence), no lack of fit of the model was perceived: calculated $F_{\text{ratio}} = 0.0319$, standard $F_{\text{ratio}} = 1.8456$, p -value = 0.9999. With a p -value > 0.10 (lack of fit statistically insignificant), the model effectively predicts the experimental response data within 95 % confidence limits (Fig. 5).

3.6. Phage virion one-step growth curve

Non-linear fitting the 4-PL model (2.10) to the phage virion growth data (r^2 (coefficient of determination) = 0.9990) revealed that phage PcgS01F feature latent and intracellular accumulation periods of 40 and 10 min, respectively (Fig. 6), with a virion morphogenesis yield (i.e., the burst size) of 46 virions/host cell.

A statistical analysis of lack of fit of the 4-PL model [45,46] to the experimental phage growth data allowed to conclude that, at $\alpha = 0.05$ (95 % confidence), no lack of fit of the model was perceived: calculated $F_{\text{ratio}} = 0.0539$, standard $F_{\text{ratio}} = 2.1306$, p -value = 1.0000. Because lack of fit is the variation of the experimental data around the fitted model, a p -value > 0.10 means that the model effectively predicts the experimental response data within 95 % confidence limits (as shown in Fig. 6).

3.7. In vitro phage-bacteria inactivation experiments

The bacterial concentration in the control (BC) increased substantially (ANOVA, $p < 0.05$) during the 720 min of incubation (as shown in Fig. 7a). When applying the phage, all MOI tested succeeded in controlling the bacteria during the first 360 min of experiment (as shown in Fig. 7a). At MOI 10, phage PcgS01F managed to reduce the bacterial load by ca. 95 % after 360 min of incubation with its host, which slightly

reduced to ca. 90 % up to 540 min of assay (Fig. 7a,b). This number slightly decreased to ca. 75 % after 720 min of incubation (Fig. 7b). When using phage PcgS01F at MOI 0.01, ca. 85 % of bacterial reduction was attained after 540 min, which quickly reduced to ca. 45 % after 720 min of incubation (ANOVA, $p > 0.05$) due to bacterial regrowth. At MOI 1000, the performance of phage PcgS01F was much better than that at MOI 10 and MOI 100 after 600 min and 720 min of incubation (ANOVA, $p < 0.05$). After 360 min of phage-bacteria contact, MOI 10 promoted decreasing bacterial reductions until the end of the assay, stabilizing at ca. 75 % bacterial reduction. On the other hand, however, the phage at MOI 1000 promoted increasing bacterial reductions until the end of the assay, stabilizing at a significant 82 % bacterial load reduction (Fig. 7b, ANOVA, $p < 0.05$). MOI 100 accompanied the bacterial reduction trend of MOI 1000 but stayed always below the reduction level attained when using the phage at MOI 1000 (Fig. 7a and Fig. 7b). At MOI 100 and 1000, the performance of phage PcgS01F was similar between 360 and 480 min and at 600 min of assay (ANOVA, $p > 0.05$) (Fig. 7b). At MOI 0.01 and 0.1, the performance of phage PcgS01F was similar between 540 and 600 min of assay (ANOVA, $p > 0.05$) (Fig. 7b). Increasing the MOI from 1 → 10 → 100 → 1000 did significantly increase the inactivation factor after 360 min of incubation (ANOVA, $p < 0.05$) (Fig. 7a and Fig. 7b), but after 600 min of incubation MOI 1 was not very effective in maintaining bacterial reduction. MOI 0.01 was the worst in terms of bacterial reduction performance (Fig. 7a and Fig. 7b), attaining only ca. 44 % (ANOVA, $p > 0.05$) of bacterial load reduction after 720 min of incubation with the host. When phage PcgS01F was added to the planktonic host cells, a significant bacterial regrowth was observed after 480 min of incubation (ANOVA, $p < 0.05$) for MOI 0.01, 0.1 and 1 (Fig. 7a). On the other hand, adding the phage at MOI 1000 succeeded in preventing bacterial regrowth (Fig. 7a) until the end of the treatment timeframe. Despite this, by the end of the incubation timeframe the bacterial densities in the different treatments using phage PcgS01F were significantly lower at all MOI than that observed for the bacterial control (BC, Fig. 7a).

3.8. Evaluation of the frequency of emergence of Pcg mutant cells resistant to the phage

The frequency of Pcg IBSBF-158 mutant cells resistant to phage

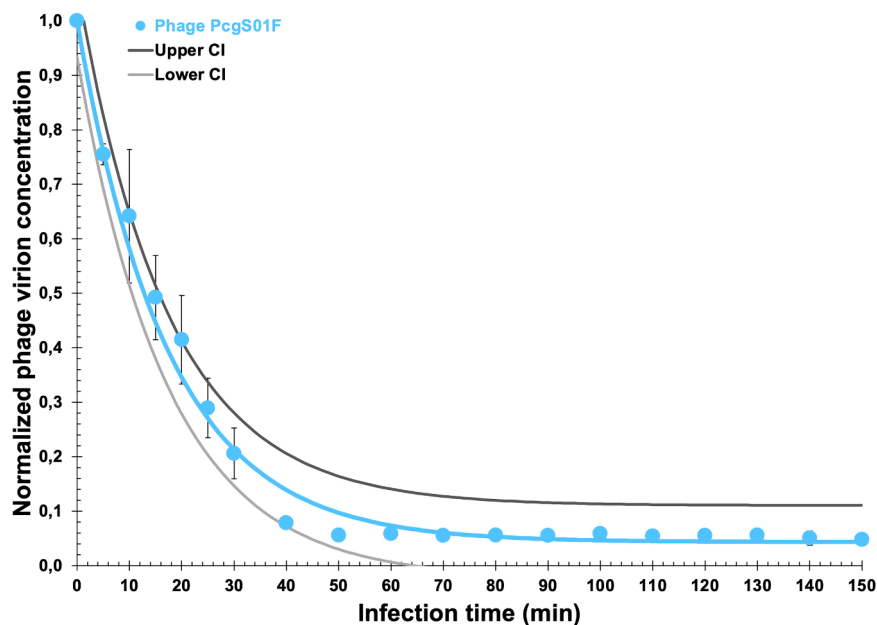


Fig. 5. Adsorption curve of phage PcgS01F onto the host (Pcg IBSBF-158) cells. The blue line represents the non-linear fitting of the normalized phage adsorption decay model to the experimental data. Values are the means of three independent assays; Error bars represent the standard deviations. The grey lines represent the upper (dark grey) and lower (light grey) 95 % confidence interval (CI) for the non-linear fitting performed.

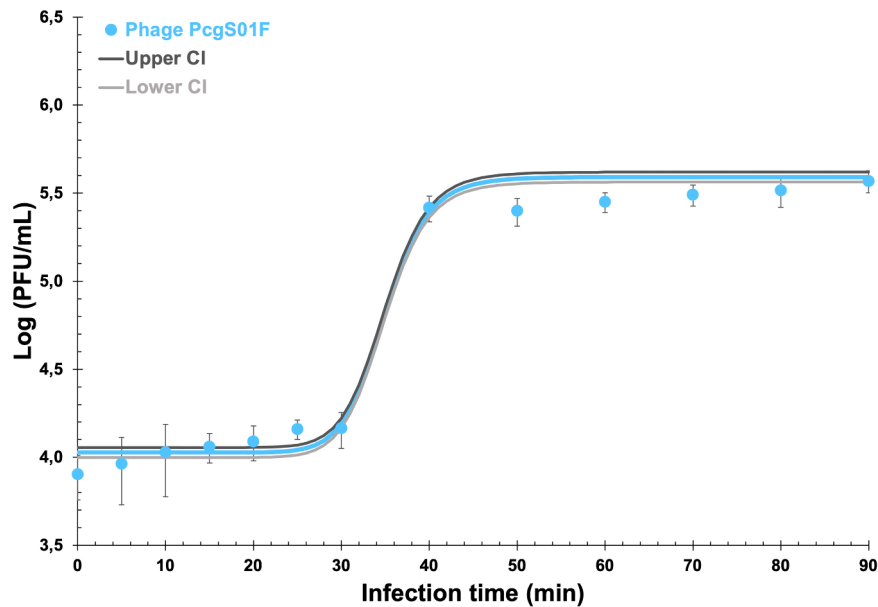


Fig. 6. One-step growth curve of phage PcgS01F within cells of the host (Pcg IBSBF-158). The blue line represents the non-linear fitting of the 4-PL model to the experimental phage growth data. Values are the means of three independent assays; Error bars represent the standard deviations. The grey lines represent the upper (dark grey) and lower (light grey) 95 % confidence interval (CI) for the non-linear fitting performed.

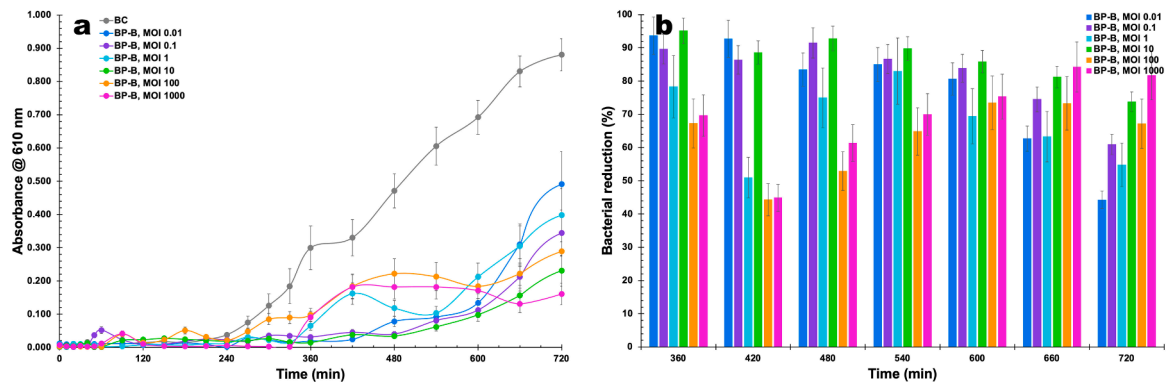


Fig. 7. *In vitro* inactivation of planktonic Pcg IBSBF-158 cells by phage PcgS01F at a multiplicity of infection (MOI) of 0.01, 0.1, 1, 10, 100 and 1000 (a), and planktonic bacterial reductions (%) attained at all MOI for phage PcgS01F (b), during a 12 h treatment timeframe. Bacterial concentration: BC, bacterial control; BP-B, bacteria with phage. Each value represents the mean of three independent assays. Error bars represent the standard deviation.

PcgS01F was 6.194×10^{-4} (0.062 %) (Table 2). After 5 h of incubation with phage PcgS01F spotted on a lawn of Pcg IBSBF-158, it was visible the appearance of the clear zone of lysis promoted by the phage virions which became more and more translucent up to 12 h of incubation (data not shown). However, after 8 h of incubation in the presence of its host, probable resistant colonies within the clear zone of lysis started to appear, which became very intense after 24 h of incubation. After this period, the number of probable resistant bacterial colonies remained constant up to 96 h of incubation. These results are consistent with the phage-bacteria inactivation assays performed, where

a significant bacterial regrowth was observed after 8 h of experiment (Fig. 7a) especially when adding the phage at lower MOI (viz. 0.01, 0.1 and 1). The use of phages to inactivate bacteria brings along the major concern that phage-resistant bacterial mutants emerge, counteracting the phage predation effect [57,91–96]. Phage PcgS01F did not fully prevent bacterial regrowth during treatment (Fig. 7a and Fig. 7b). During the bacterial predation experiments (Fig. 7a) it is not possible to know for sure whether the surviving bacteria are really resistant to the phage, simply because the bacterial cells are collected from liquid TSB samples containing both phage virions and planktonic cells and it is impossible to guarantee that all the bacterial cells that were tested effectively contacted with the virions via random encounters. The frequency of spontaneous emergence of phage-resistant mutant cells, determined only for those cells that were in contact with the phage virions, was limited (6.194×10^{-4}) (Table 2).

Table 2			
Frequency of emergence of Pcg IBSBF-158 spontaneous resistant mutants to phage PcgS01F.			
Phage	Control Sample (CFU/mL)	Sample Treated with Phage (CFU/mL)	Frequency of Phage-Resistant Bacterial Mutants
PcgS01F	(1.430 ± 0.186) × 10 ¹⁰	(8.860 ± 1.321) × 10 ⁶	6.194 × 10 ⁻⁴

3.9. Stability to abiotic factors

3.9.1. pH experiments

Testing the impact of different pH values (5.0, 7.0, and 10.0) upon the lytic viability of phage PcgS01F allowed one to observe that the concentration of lytic viability decreased at both acidic and basic pH (Fig. 8a); however, the differences among pH values 5.0 and 10.0 were not statistically significant (Fig. 8a, ANOVA, $p > 0.05$) up to 4 h of incubation, after which phage PcgS01F lytic viability decreased more at

pH 10.0 than at pH 5.0 up to 12 h of incubation (Fig. 8a, ANOVA, $p < 0.05$). At pH 5.0, phage PcgS01F lost ca. 0.6 log PFU/mL, whereas at pH 10.0 it lost 0.5 log PFU/mL, over a 72 h period of incubation, differences that were not statistically significant (Fig. 8a, ANOVA, $p > 0.05$). At pH 7.0, phage PcgS01F lost a little amount of lytic viability during the first 2 h of incubation (viz. ca. 0.2 log PFU/mL) but after this initial period it endured until 12 h of incubation without any detectable loss of activity (Fig. 8a, ANOVA, $p > 0.05$), after which it lost only ca. 0.3 log PFU/mL until the end of the 72 h timeframe of assay. Hence, phage PcgS01F revealed to be quite resistant to the effects of pH upon its lytic viability.

3.9.2. Temperature experiments

Testing the effects of temperature upon phage PcgS01F lytic viability revealed that this phage was very stable at 4 °C, 17 °C and 25 °C, without any loss of activity at these temperatures during the first 48 h of incubation (Fig. 8b, ANOVA, $p > 0.05$). After this timeframe, a minor decrease of lytic activity was noticed (viz. 0.4 log PFU/mL) up to 72 h of incubation (Fig. 8b). At 40 °C, the same trend of virion lytic stability was observed until the first 12 h of incubation (Fig. 8b). After this time period, however, phage PcgS01F lost ca. 1 log PFU/mL of lytic viability. Hence, after 24 h of incubation, the reduction in the concentration of viable phage virions was much higher at 40 °C than at 4 °C, 17 °C or 25 °C (Fig. 8b, ANOVA, $p < 0.05$).

3.9.3. Solar radiation experiments

When phage PcgS01F was exposed to solar radiation, the abundance of lytic viability endured throughout the exposure timeframe, with only a marginal difference between control and test (Fig. 8c, ANOVA, $p > 0.05$). The concentration of viable lytic virions (SR) decreased by only ca. 0.2 log PFU/mL (Fig. 8c, ANOVA, $p > 0.05$) after 7 h of exposure, when compared with the phage control (SR-C). Phage PcgS01F revealed, therefore, to be quite resilient when exposed to direct sunlight.

3.10. Genomic characterization

The genome of phage PcgS01F was sequenced and duly assembled, resulting in a contig of 44,679 bp. Circularization of the contig was possible with ccfind, with overlapping of both end sequences, and the assembled phage genome is complete. The GC content of the PcgS01F phage genome is 44.42 %.The overall features of phage PcgS01F genome assembly and annotation are summarized in Table 3.

The genome of phage PcgS01F encodes 2 tRNAs and 76 protein coding sequences (CDS) (Supplementary Table S1). Comparison of the annotated CDS in the genome of phage PcgS01F with different databases unveiled that 47 of them are predicted as hypothetical proteins or proteins of unknown function (Supplementary Table S1). Typical structural proteins such as capsid (head), neck, tail, and tail fiber were annotated

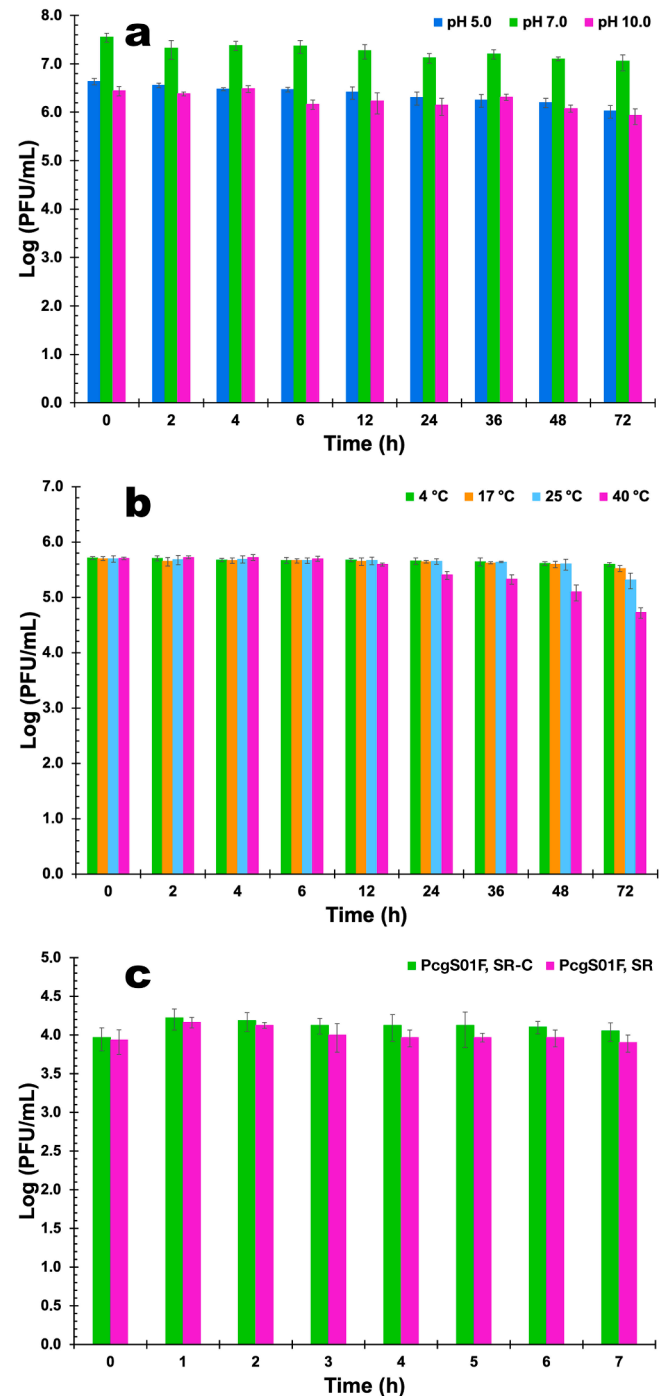


Fig. 8. Survival of phage PcgS01F following exposure to different pH (a), different temperature (b), and solar radiation (c). Values represent the mean of three independent experiments; error bars represent the standard deviation. SR: phage exposed to solar radiation; SR-C: phage control - phage not exposed to solar radiation.

Table 3
Genomic features of phage PcgS01F genome.

Feature	Phage PcgS01F
NCBI/Genbank accession number	PP134917
Genome size	44,679 bp
Number of PE reads mapping in the final assembly	42,771,707 (94.58 %)
Average sequencing coverage (calculated as (length of reads × number of reads mapping) / genome size)	281,518.4 ×
GC content	44.42 %
tRNA genes	2
Protein-coding genes (CDS) predicted	76
• With function assigned	29 (38.2 %)
• Hypothetical / unknown function	47 (61.8 %)
Similar phage genome sequences	Escherichia phage
NCBI/RefSeq accession number of similar phage	vB_EcoS_CEB_EC3a KY398841
Morphotype	Siphovirus
Family / Genus	Drexlerviridae / Guelphvirus

in phage PcgS01F genome, along with DNA-repair and metabolism-related proteins and host lysis proteins (endolysin, spanin), and tail length tape measure protein (dictating tail length and facilitating DNA translocation into the host cell cytoplasm during infection) (Supplementary Table S1). A circular map of the annotated genome of phage PcgS01F is displayed in Fig. 9.

Using the web-based ViPTree application[79], relationships and similarities between phage PcgS01F and other prokaryotic dsDNA viruses were analyzed [43,97] as described by Silva et al. [43] and Xuan et al. [97]. Fig. 10 displays the viral proteomic tree that resulted from ViPTree analysis of phage PcgS01F and related phages.

Results showed that phage PcgS01F was grouped into one small group with other 7 phages (Fig. 10), with all such phages belonging to the *Drexlerviridae* family according to the updated ICTV taxonomic classification.

On the network analysis, phage PcgS01F was grouped with 7 phages at the family level (Fig. 10b), viz. vB_EcoS-IME253, Rtp, vB_EcoS-IME542, vB_EcoS-1221OI, DTL, vB_EcoS-ACG-M12, and vB_EcoS-CEB_EC3a. Here we can see that phage PcgS01F and phage vB_EcoS-CEB_EC3a (the phage closest to phage PcgS01F) are alone on their branch, showing high phylogenetic similarity (Fig. 10b). According to the analyzes carried out using the ViPTree application, tBLASTx genomic sequence comparison of phage PcgS01F with other phage genomes deposited in the Virus-Host DB (<https://www.genome.jp/virushostdb>) [98] showed high similarity of phage PcgS01F with another phage of the *Drexlerviridae* family: Escherichia phage vB_EcoS-CEB_EC3a (93.2 %; NC_047812) (Fig. 11).

3.11. DNA structural features of phage PcgS01F

Studying the mechanical properties of the genome of this siphovirus

was essential for a better and deeper understanding of the differences in the infection rates of its target bacterial cells when compared with results from similar analysis of two myoviruses lytic for the same phytopathogen [43], with concomitant very different yields of virion morphogenesis within infected cells (viz. 46 virions per host cell (phage PcgS01F) versus 124 virions per host cell (phage PsgM02F [43]) versus 12 virions per host cell (phage PsgM04F [43])).

Fig. 12 displays the results from DNA shape (Propeller Twist (ProT), Minor Groove Width (MGW), Roll and Helical Twist (HelT)) calculations for the assembled phage genome.

The interaction between proteins and DNA plays a pivotal role in various biological regulatory processes, with DNA-protein interactions relying on the inherent mechanical properties of DNA, allowing for deformation, bending, and twisting [99]. This process is significantly influenced by the DNA shape [100], encompassing physical and geometrical properties like the MGW, HelT, ProT, and roll angle.

The HelT is defined as the angular rotation required to transition from one base pair to the next [101,102]. The minor groove of DNA plays an important role in the recognition by various proteins [103,104]. ProT involves a rotation about the long axis, causing two bases of a pair to be non-coplanar [105], while the roll degree serves as a local angular parameter indicating the bending into the grooves of DNA [106]. These features contribute, to some extent, to DNA bendability, and offer insights into the interaction dynamics between proteins and DNA.

Genome mechanical features can now be predicted based on the DNA sequence, via using high-throughput methods, revealing inherent preferred conformations [107,108].

In this study, the DNA shape features of the PcgS01F phage genome were predicted using a high-throughput method called DNAShape (a web-based application), with the outcomes being visually represented as

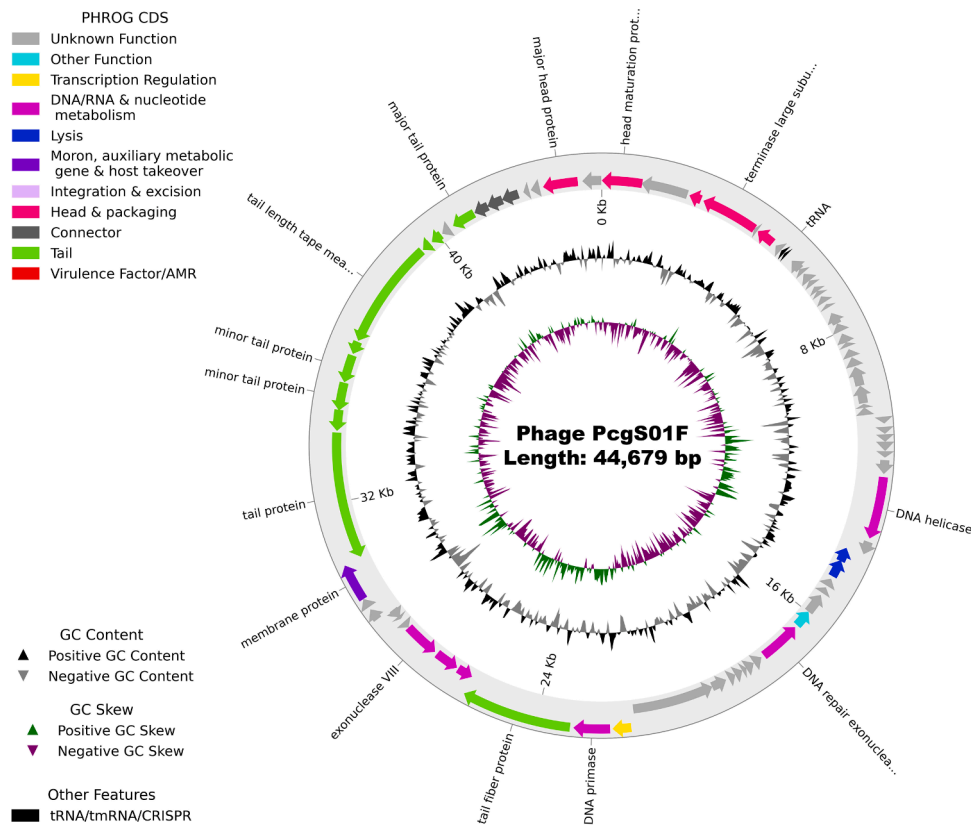


Fig. 9. Annotated genome map of phage PcgS01F, displaying GC skew, G+C content and predicted coding sequences (CDSs). The coloured (except gray) arrows in the outer ring represent the annotated CDS according to the annotation in Supplementary Table S1 whereas the gray arrows correspond to hypothetical proteins, and black arrows correspond to tRNAs. The arrows represent the direction of transcription (strand + or -).

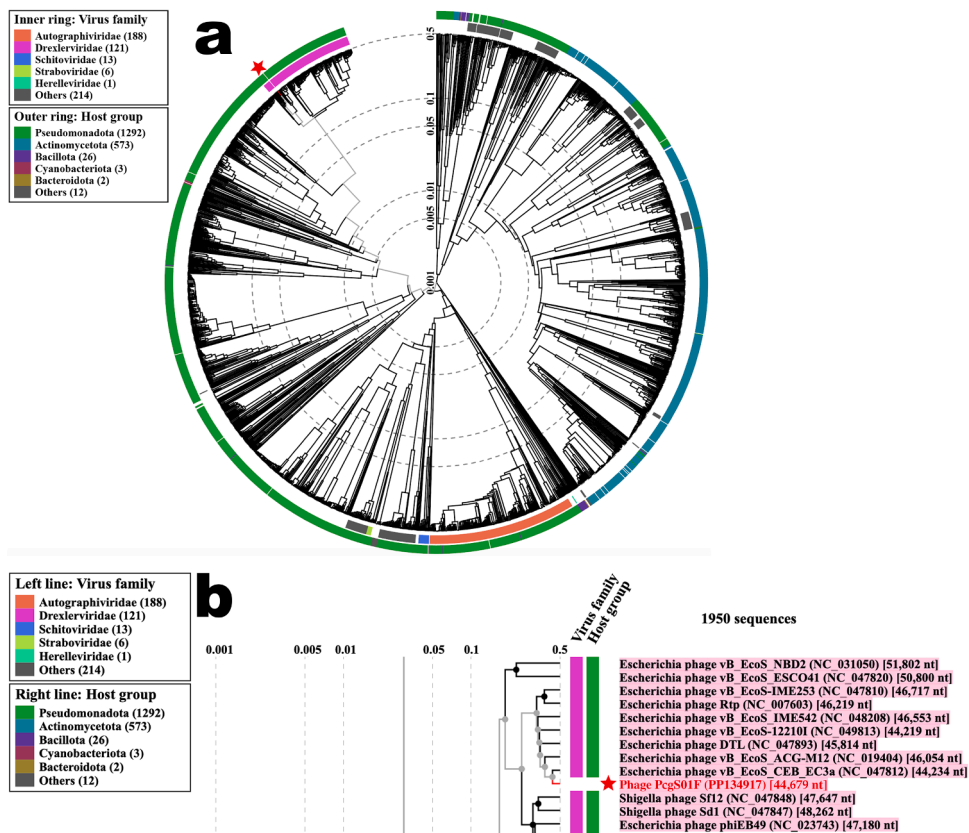


Fig. 10. Viral proteomic tree from ViPTree analysis of phage PcgS01F and related phages. (a) 5633 phage genomes were used as reference sequences to build phylogenetic trees using ViPTree. In this Figure, phages are identified according to their official ICTV classification, with the inner and outer rings indicating their virus family and host group, respectively. (b) Expanded view of the region of the tree containing the most closely related phages. The branch containing phage PcgS01F is displayed in red. The red star pinpoints the location of phage PcgS01F.

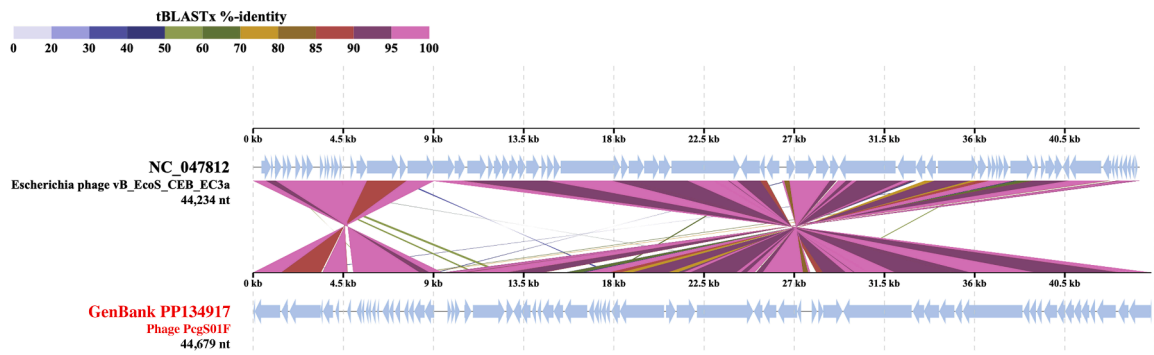


Fig. 11. Genome sequence alignment and comparison of phage PcgS01F with another *Drexlerviridae* phage genome exhibiting co-linearity detected by TBLASTX using the ViPTree server [79]. Homologous regions detected by a TBLASTX search are connected by segments colored based on amino acid identity. The upper color bar shows the % identity of TBLASTX.

heatmaps (Fig. 12). ProT refers to the angle between the planes of two DNA bases. When the nearer base rotates clockwise, the angle is considered positive; counterclockwise rotation yields a negative angle. This measurement reflects DNA's flexibility, with a larger ProT indicating a more rigid DNA structure [105]. The presence of propeller twists in base pairs complicates their stacking into dinucleotide steps compared to in-plane base pairs. Specifically, it creates a chemical locking effect, increasing the rigidity of dinucleotide steps with high ProT angles. Therefore, smaller ProT facilitate easier sliding of base pairs over each other within the constraints of the DNA backbone [105]. In the ProT heatmaps (Fig. 12), it is observed higher and lower ProT regions within the phage genome sequence. The higher and lower values

are represented by yellow and dark brown colors, respectively (Fig. 12, ProT). Notably, higher average ProT angles are observed in the PcgS01F phage genome (Fig. 12, ProT), with a mean of -7.29° , in contrast to the values reported by Silva et al. [43] for two myoviruses (-8.21 for phage PsgM02F and -7.56 for phage PsgM04F), suggesting that on average there are more flexible regions in PcgS01F genome than in the two myoviruses genomes. However, while ProT values may be comparable on average across those phage genomes, variations can arise at the local level due to differences in the distribution of ProT within each genome. The DNA helical structure arises from the stacking of individual bases on top of each other, with the bases twisting slightly to optimize their hydrophobic interactions, thereby forming the helix [109]. The

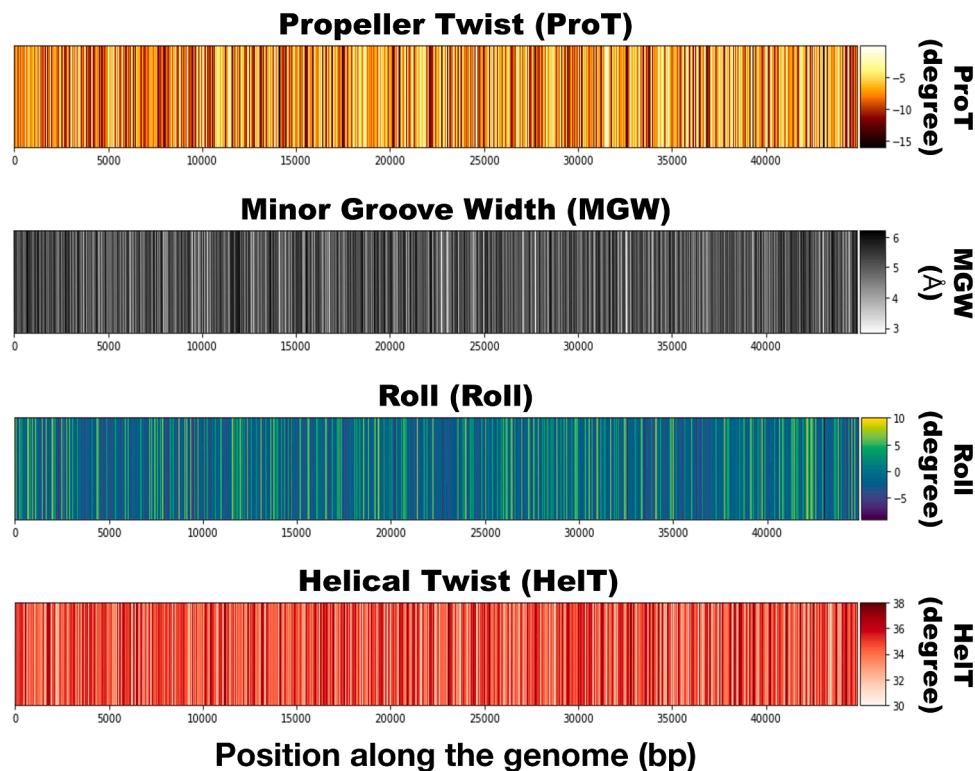


Fig. 12. Heatmap patterns from the results of DNA shape (Propeller Twist, Minor Groove Width, Roll and Helical Twist) calculations for the assembled phage PcgS01F genome.

flexibility of DNA twist plays a crucial role in its various biological functions. Previous studies have examined this flexibility using short-base sequences due to their looping ability [107,110], as well as investigating the bending and twisting deformations necessary for protein-DNA interactions [111–113]. Collectively, these findings suggest that HelT serves as an effective metric for DNA flexibility. The evaluation of HelT heatmap in PcgS01F phage genome revealed a high HelT degree (Fig. 12, HelT), with a mean value of 34.435° , comparable to the values reported by Silva et al. [43] for two myoviruses (-34.61° for phage PsgM02F and 34.49° for phage PsgM04F). The HelT heatmap provide insights into the distribution of HelT angles in the phage genome, with dark red indicating a higher angle and white representing a lower HelT angle (Fig. 12: HelT).

The roll angle serves as a local angular parameter delineating the bending and twisting of adjacent base pairs [114,115]. Notably, significant positive roll values in base pair steps indicate weak stacking interactions [107]. Tolstorukov et al. [106] proposed a correlation between roll and conformational flexibility in the nucleosome. Through molecular dynamic simulations, those researchers demonstrated that setting the roll value to zero at all base-pair steps nearly straightens the DNA molecule, illustrating that those variations in roll accounted for DNA bending. Furthermore, they observed that negative roll values cause the DNA to bend towards the minor groove width, while positive roll values induce bending towards the major groove width. These findings underscore the relationship between roll and DNA bendability.

The Roll angle values of the PcgS01F phage genome are displayed in the heatmap with a yellow color, indicating a higher degree, and a dark blue color indicating a lower degree of Roll (Fig. 12: Roll). Indeed, we can discern variations in the distribution of roll angles in the PcgS01F phage genome. The average Roll value for the PcgS01F phage genome is -0.769° , comparable to the values reported by Silva et al. [43] for two myoviruses (-0.89° for phage PsgM02F and -0.82° for phage PsgM04F). The roll angles for these phage genomes are negative, with a minor difference. A less negative roll angle value for the PcgS01F genome

might indicate a lower degree of bending towards the minor groove width in certain regions of the phage genome when compared with the PsgM02F genome [43], but more studies are needed to prove this statement.

The minor groove width (MGW) varies geometrically with sequence and can significantly influence DNA shape recognition by proteins [116]. A narrower minor groove suggests greater accessibility of base edges to proteins, such as transcription factors [107]. This might be due to the groove width compression caused by the unusually close spacing between recognition helices of each helix-turn-helix motif [117]. On the other hand, a study conducted by Oguey et al. [118] demonstrated a relationship between flexibility and MGW, suggesting that this relationship is likely an intrinsic property of DNA. The MGW can be also visualized in the respective heatmap (Fig. 12: MGW), where black indicates a higher MGW angle value, while white indicates a lower MGW angle value. The distribution of black and white patterns varies along the phage genome sequence, highlighting the varying significance of MGW in different regions of the phage genome at the local level. The mean MGW angle value for the PcgS01F phage genome is 5.040° , comparable to the values reported by Silva et al. [43] for two myoviruses (4.98° for phage PsgM02F and 5.02° for phage PsgM04F). The PcgS01F phage genome exhibits a wider and less concentrated MGW value compared to those values of phages PsgM02F and PsgM04F reported by Silva et al. [43]. The less heightened narrowness of the MGW in the PcgS01F phage genome may suggest decreased protein-DNA interactions and less bending in certain regions of the genome, as proteins like transcription factors tend to induce DNA bending, as opposed to phage PsgM02F [43]. A discernible difference in the MGW between the phage genomes can be noticed, which might indicate a difference in MGW in the PcgS01F, PsgM02F and PsgM04F genomes at nucleotide and global level, with a narrower MGW for the PsgM02F genome.

After analyzing the DNA shape of the PcgS01F phage genome, a Pearson correlation analysis was conducted on its DNA mechanical features, to explore the linear relationship among the DNA mechanical

features. The results were visually represented in two graphs, displaying the Pearson correlation values for each structural feature (Fig. 13).

The Pearson correlation analysis between the four predicted structural features of the genome of phage PcgS01F revealed both negative and positive correlations. Notably, a positive correlation was observed between the Roll angle degree and the MGW, with a value of 0.51 for the PcgS01F phage genome. This relationship aligns with previous research by Fratini et al. [109], which demonstrated that the roll angle measures the degree to which successive base pairs open towards the minor groove. Consequently, higher positive roll angles may yield wider minor groove widths. Conversely, a negative correlation was observed between the HelT and the MGW, with a value of -0.22° . Liebl et al. [119] demonstrated that DNA overwinding decreases the MGW, leading to an increased HelT due to enhanced stacking distances between adjacent base pairs. Thus, we can infer a probable negative correlation between HelT and MGW.

Moreover, we observed positive and negative correlations among DNA structural features, such as the positive correlation between Roll and HelT, which measured 0.023 (Fig. 13). Although the Pearson correlation, in this case, is close to zero, indicating a minimal association between these features, prior research has identified an anti-correlation likely due to the presence of specific pyrimidine/purine steps in the genome [120–122].

Furthermore, a negative correlation between ProT and HelT (viz. -0.24) was observed in the phage genome (Fig. 13). Propeller twisting aims to optimize stacking interactions. When ProT values approach 36 degrees, they tend towards zero, indicating optimal stacking

interactions. Deviations might result in diminished stacking. El Hassan and Calladine [105] noted that more negative ProT angles in double-stranded DNA (dsDNA) are associated with high flexibility, whereas ProT angles ranging from -1 to -3 degrees are linked to greater rigidity in dsDNA. This statement is supported by Basu et al. [82], who also showed that high ProT correlates with low cyclizability, indicative of reduced DNA bendability.

Based on these findings, we can infer that the PcgS01F phage genome shows a degree of correlation among the DNA structural features which might influence the mechanical properties of the DNA. DNA shape could potentially impact the biology of the phages. Narrow MGW, distinctive HelT conformations, alterations in roll, and variations in ProT may collectively affect various biological processes associated with the phages, such as DNA translocation. This could happen by facilitating enhanced DNA-protein interactions, improving accessibility, and contributing to the stability of DNA structure during the translocation event.

While our findings suggest a difference between the PcgS01F phage genome and the myovirus phage genomes described by Silva et al. [43], it is important to note that these analyses were primarily designed to study the impact at the local level rather than at the global level. Therefore, further studies are necessary to validate these observations and understand their broader implications.

A custom Python script was then used to investigate the occurrence of the possible 16 dinucleotides (AA, AC, AG, AT, CA, CC, CG, CT, GA, GC, GG, GT, TA, TC, TG, and TT) in the genome of phage PcgS01F. This was accomplished by counting the occurrences of each dinucleotide in

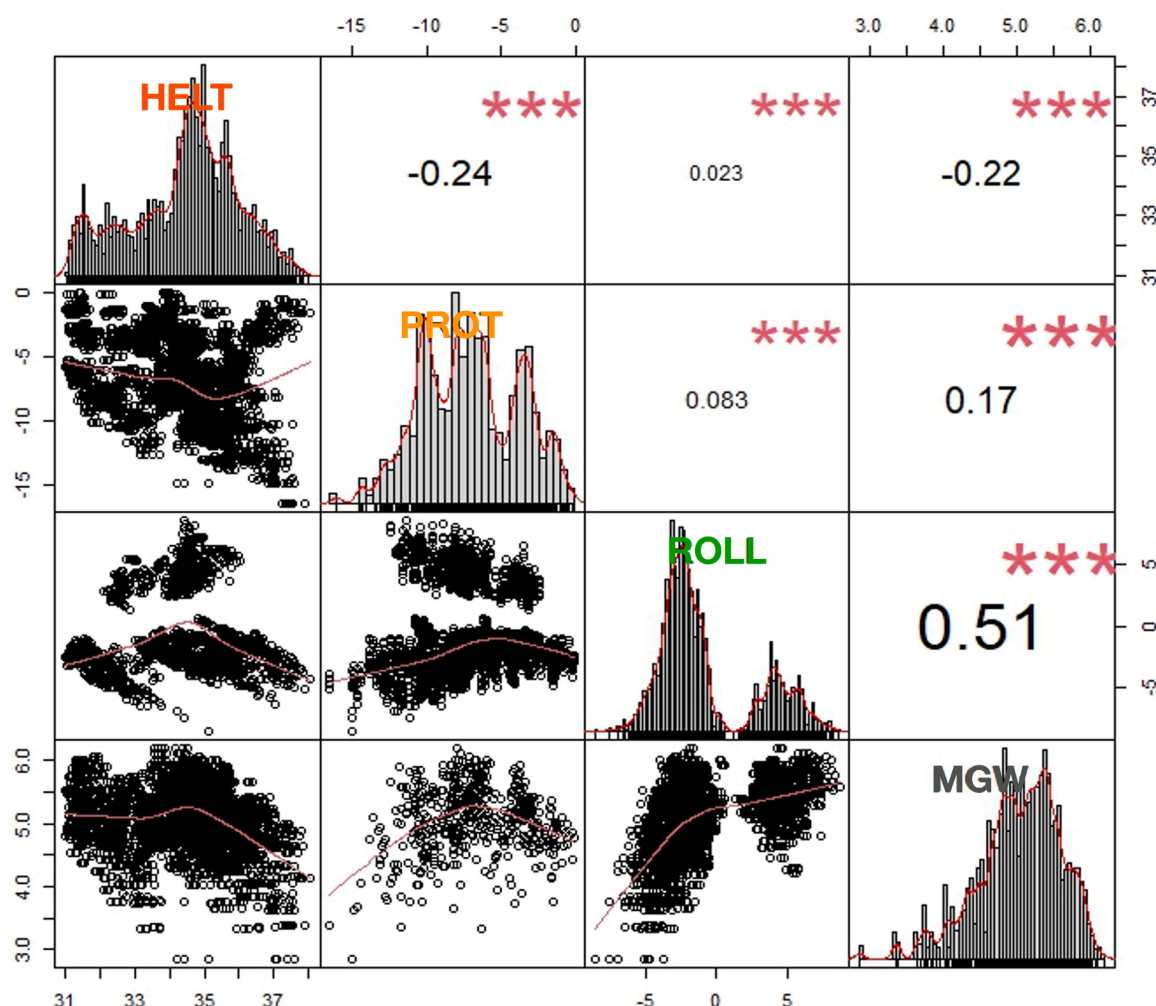


Fig. 13. Pearson correlation analysis between the 4 predicted structural features of the genome of phage PcgS01F.

the phage genome and dividing them by the genome's length. Fig. 14 displays the dinucleotide frequency in the genome of phage PcgS01F.

The frequency of occurrence of dinucleotides (AA, AC, AG, AT, CA, CC, CG, CT, GA, GC, GT, TA, TC, TG, and TT) in the PcgS01F phage genome was calculated and displayed in a bar graph (Fig. 14), exhibiting the percentage of each dinucleotide in the genome. Notably, certain dinucleotides exhibit higher frequencies in the phage genome. Dinucleotides AT, CA, CT, GC, TA, TC, TG, and TT demonstrate higher frequencies (Fig. 14). Conversely, the dinucleotides CC, CG, GA, and GG exhibit lower frequencies. The AT dinucleotide exhibits the highest occurrence frequency in the PcgS01F genome, reaching 6.96 %, followed closely by the CA dinucleotide with the second-highest occurrence frequency at 6.41 % (Fig. 14). In contrast, the GG dinucleotide displays the lowest occurrence frequency, accounting for 2.97 %.

In Fig. 15, the predicted intrinsic cyclizability along the phage genome at 7 bp resolution can be observed for phage PcgS01F, following the procedures detailed by Basu et al. [82]. Their approach involved the simultaneous measurement of intrinsic cyclizability across a broad spectrum of 90,000 distinct 50 base pair DNA sequences.

Notably, the cyclizability pattern vary along the phage genome (Fig. 15a,c), revealing distinct peaks of cyclizability. In the PcgS01F phage genome, there are some peaks reaching a maximum value of 0.712 and a minimum of -0.785 . However, despite the observed higher cyclizability peaks in the PcgS01F phage genome, the mean cyclizability value for this genome is -0.14319 , a value that lies between those reported by Silva et al. [43] for two myoviruses (with an average of -0.12567 for the PsgM02F phage genome and -0.16482 for the PsgM04F phage genome). This suggests that, on average, the PcgS01F phage genome has lower cyclizability compared to the PsgM02F phage genome but higher cyclizability compared to the PsgM04F phage genome. Overall, these results might suggest that the PcgS01F phage genome is somewhat more flexible than the PsgM04F phage genome but somewhat more rigid than the PsgM02F phage genome.

Fig. 15b allows to observe the distribution of cyclizability values' sizes and frequency of apparition in the PcgS01F phage genome. As we can see, the distribution of cyclizability from the PcgS01F genome goes from ca. -0.80 – 0.60 . Based on this distribution we can suggest that there are more peaks with a high cyclizability (close to zero) in the PcgS01F phage genome.

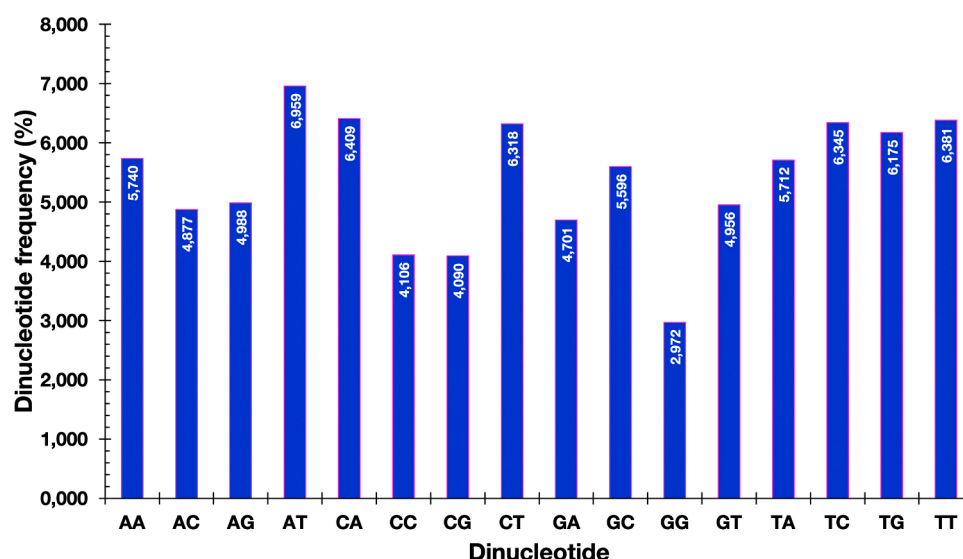


Fig. 14. Dinucleotide frequency in the genome of phage PcgS01F.

3.12. Ex vivo phage-bacteria inactivation experiments in artificially contaminated coffee plant leaves

Pcg density in the BC increased by 4.02 log CFU/mL (as shown in Fig. 16a) during the 36 h of incubation. The ANOVA revealed that the differences between sampling times among treatments of Pcg with phage PcgS01F ($p < 0.05$) and the differences between viable Pcg bacterial density among treatments ($p < 0.05$) were significant when comparing the two MOI tested. Phage PcgS01F effectively infected and killed Pcg cells in coffee plant leaves artificially contaminated. The maximum of Pcg inactivation was a reduction of 2.86 and 4.52 log CFU/mL in relation to the uninfected control, for MOI 10 and 1000, respectively (Fig. 16a, Bonferroni, $p < 0.05$), after 6 h of incubation, whereas the maximum of Pcg inactivation was 2.83 and 6.45 log CFU/mL, for MOI 10 and 1000, respectively (Fig. 16a, Bonferroni, $p < 0.05$), after 24 h of incubation, and 3.05 and 7.66 log CFU/mL, for MOI 10 and 1000, respectively (Fig. 16a, Bonferroni, $p < 0.05$), after 36 h of incubation. MOI 1000 promoted the highest reduction in Pcg counts after 3 h, 6 h, 12 h, 24 h and 36 h of incubation. After 12 h of incubation, the Pcg cell counts (reduction of 2.87 and 5.98 log CFU/mL, for MOI 10 and MOI 1000, respectively) were significantly different (Bonferroni, $p < 0.05$) when compared to the bacterial control (Fig. 16a). After the first 6 h of treatment, a slight bacterial regrowth was observed until the end of the incubation timeframe (ANOVA, $p < 0.05$) (Fig. 16a) for MOI 10, whereas no bacterial regrowth whatsoever was observed for the phage treatment at MOI 1000 during the whole treatment timeframe. While phage controls (PC) remained constant throughout the experiments (ANOVA, $p > 0.05$), higher initial phage PcgS01F titres (i.e., higher MOI) did not result in higher virion concentrations (due to phage replication) towards the end of the incubation timeframe relative to their initial counterparts, with no statistical difference being observed (3.94 and 3.94 log PFU/mL, Bonferroni, $p > 0.05$, for MOI 10 and 1000, respectively) (Fig. 16b).

4. Discussion

Developing new, environmentally-friendly and bacterial resistance-avoiding, biocontrol strategies for controlling infections by the phytopathogen *Pseudomonas coronafaciens* pv. *garcae* (Pcg) in coffee plantations, is a challenge. In the research work entertained herein, a novel isolated lytic bacteriophage for Pcg (viz. PcgS01F) was characterized from physicochemical, biological and genomic points of view, and duly tested in *in vitro* (planktonic) and *ex vivo* Pcg inactivation experiments.

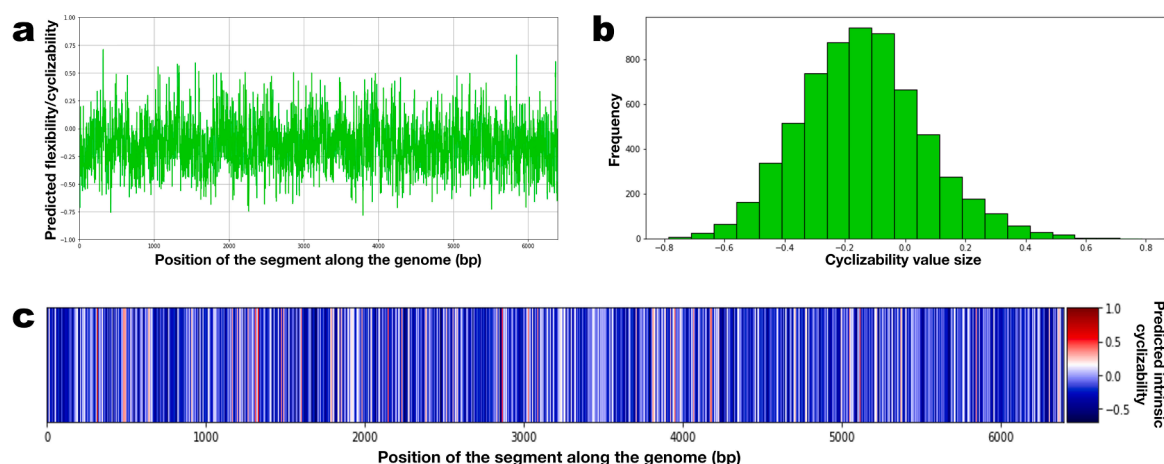


Fig. 15. Predicted intrinsic cyclizability along the phage genomes at 7 bp resolution (a), distribution of cyclizability values and frequency of apparition in the genome (b), and heatmap pattern of the cyclizability values of the genome of phage PcgS01F.

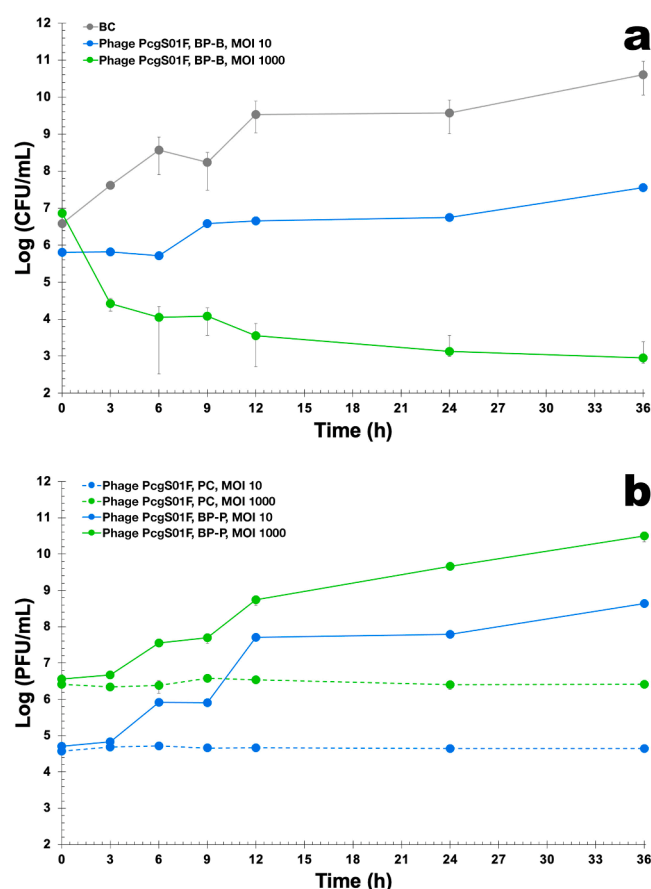


Fig. 16. Ex vivo inactivation of Pcg IBSBF-158 in artificially contaminated coffee plant leaves by phage PcgS01F at MOI values of 10 and 1000, during a 36 h assay. (a) Bacterial concentration: BC, bacterial control; BP-B, bacteria with phage; (b) Phage concentration: PC, phage control; BP-P, bacteria with phage. Values represent the mean of three independent assays whereas error bars represent the standard deviation.

Phage PcgS01F formed clear plaques of lysis on the host lawn, without secondary halo, with diameters of ≈ 1.5 mm (as shown in Fig. 3a) and was identified as belonging to the class Caudoviricetes with siphovirus-like morphotype (as shown in Fig. 3d,e). The novel isolated phage presented a molar extinction coefficient (Fig. 4b) of the same order of magnitude as the ones determined by Silva et al. [33] for two

myoviruses lytic for the same phytopathogen.

Structural proteins (such as capsid, tail, baseplate, and spike) making up bacteriophage virions absorb ultraviolet radiation in the range 278–280 nm mainly because of a few amino acid chromophores: side chains of tryptophan, tyrosine and phenylalanine, and disulfide bonds of cysteine. Attenuation of radiation that reaches the detector due to absorption by protein chromophores as well as scattering, subtracted from attenuation of radiation due to protein chromophores only, produces the characteristic absorbance spectrum for a given phage virion [33,45,46,123]. Hence, the absorption spectrum of a phage virion in the ultraviolet wavelength results solely from its structural proteins. Radiation absorption at 320 nm results almost entirely from scattering, since virion structural proteins' chromophores virtually do not absorb radiation at this wavelength. Thus, $Abs_{320\text{ nm}}$ was used to make a raw correction for light scattering from phage virions and non-virion contaminants [33,45,46,55,124].

Beyond the isolation host, novel isolated phages will most likely infect other target bacterial cells surface-decorated with the same general type(s) of receptors as the isolation host [125]. Phage PcgS01F was able to inactivate two other bacterial species (Table 1) in the spot tests performed. This was the case for *E. coli* ATCC 25922 and Psa CRA-FRU 14.10, confirmed by the efficiency of plating on these hosts (10.73 % and 0.0061 %, respectively) (Table 1). For Psa CRA-FRU 14.10, we speculate a phenomenon of low efficiency lysis from within [47] caused most likely by an abortive infection as a potential rationale for phage PcgS01F killing this bacterial strain. Bacterial resistance mechanisms to evade phage infection, such as CRISPR [126], might have also played a role for the low EOP in Psa CRA-FRU 14.10. On the other hand, phage PcgS01F was able to productively infect *E. coli* ATCC 25922 with an EOP of ca. 11 %. It is well known that phage progeny virion morphogenesis within host cells is highly dependent on both the metabolic machinery and precursor molecules from the host itself. Such draw of vital resources from the host cell imposed by the virion morphogenesis process significantly disrupts the highly interconnected network of metabolic pathways within the bacterial host [127], leading to a putative incompatibility of the host metabolic machinery in certain bacterial species for the phage virion morphogenesis. The very low EOP value and the absence of lysis observed for the large majority of the bacterial strains tested may also be explained by either the phage recognition process of specific receptors on the surface of the bacterial outer membrane or, as already mentioned, resistance mechanisms (such as CRISPR/cas) developed by bacteria to counteract phage infection [126]. Hence, the host range of the isolated phage is apparently quite narrow; this means that, in the future, isolation of new lytic phages for Pcg is highly desirable aiming at producing a cocktail with broader spectrum of activity to

control a large(r) number of Pcg strains.

Infection of a susceptible bacterial cell by a phage virion initiates with interaction and anchoring of the virion onto suitable receptors decorating the bacterial cell surface [45–47,53,128–130], following Brownian motion-based random encounters between free virions and bacterial cells together with three-dimensional architectural plasticity variations in the receptor protein conformation [32,45–47,54,131]. The adsorption rate (as shown in Fig. 5) of phage PcgS01F virions ($\delta_{\text{PcgS01F}} = 2.755 \times 10^{-10} \text{ CFU}^{-1} \text{ mL min}^{-1}$) onto Pcg IBSBF-158 cells was lower than those of phages ph002F and ph004F isolated from Silva et al. [33], onto the very same host, but of the same order of magnitude, viz. $\delta_{\text{ph002F}} = 5.204 \times 10^{-10} \text{ CFU}^{-1} \text{ mL min}^{-1}$ and $\delta_{\text{ph004F}} = 9.894 \times 10^{-10} \text{ CFU}^{-1} \text{ mL min}^{-1}$, and of the same order of magnitude of *Pseudomonas syringae* and phage phi6 ($\delta = 3.3 \times 10^{-10} \text{ CFU}^{-1} \text{ mL min}^{-1}$) [132], Psa CRA-FRU 14.10 and phage phi6 ($\delta = (1.832 \pm 0.230) \times 10^{-10} \text{ CFU}^{-1} \text{ mL min}^{-1}$) [36], *P. aeruginosa* and phage ph0031 ($\delta = 7.00 \times 10^{-10} \text{ CFU}^{-1} \text{ mL min}^{-1}$) [47], *P. aeruginosa* and phage ph0034 ($\delta = 4.00 \times 10^{-10} \text{ CFU}^{-1} \text{ mL min}^{-1}$) [47] and *P. aeruginosa* and phage ph0041x ($\delta = 2.90 \times 10^{-10} \text{ CFU}^{-1} \text{ mL min}^{-1}$) [81], but one order of magnitude higher than results published by Pinheiro et al. [36] for Psa CRA-FRU 12.54 and phage phi6 ($\delta = (7.971 \pm 1.401) \times 10^{-11} \text{ CFU}^{-1} \text{ mL min}^{-1}$), and two orders of magnitude higher than results published by Pinheiro et al. [35] for *P. syringae* pv. *syringae* van Hall 1902 (DSM 21482) and phage phi6 ($\delta = (9.495 \pm 0.660) \times 10^{-12} \text{ CFU}^{-1} \text{ mL min}^{-1}$).

The (first order) phage desorption rate from reversible phage-bacteria complexes ($\phi_{\text{PcgS01F}} = 2.500 \times 10^{-3} \text{ min}^{-1}$) was, however, lower but of the same order of magnitude as those desorption rates of phages ph002F and ph004F isolated by Silva et al. [33], onto the very same host, viz. $\phi_{\text{ph002F}} = 9.136 \times 10^{-3} \text{ min}^{-1}$ and $\phi_{\text{ph004F}} = 3.227 \times 10^{-3} \text{ min}^{-1}$, meaning that more phage PcgS01F virions remained attached to their host cells. Because anchoring of phage virions onto specific cell surface receptors is mandatory for infection of susceptible bacterial cells to occur with concomitant virion morphogenesis [81] and cell burst, the lower desorption rate for phage PcgS01F is in clear agreement with the much higher bacterial inactivation produced by this phage at longer treatment timeframes (Figs. 7a and 7b) than those achieved with phages ph002F and ph004F [33]. Consequently, at MOI 10, 100 and 1000, phage PcgS01F managed to prevent Pcg IBSBF-158 growth during the first 6 h of treatment and promoted a significant decrease in the concentration of Pcg IBSBF-158 cells after this time period when compared with the non-treated bacterial control (Fig. 7a).

When inspecting the phage virion growth data displayed in Fig. 6, a one step of growth can be clearly observed with a nice rise for the first 30–40 min (as shown in Fig. 6), levelling off afterwards. The growth characteristics of the novel isolated phage showed a relatively low burst size, viz. 46 virions / host cell, replicating well within Pcg IBSBF-158 with a short latency period (40 min).

Despite several studies suggesting that using phages with large virion morphogenesis yields and short lytic cycles do increase the efficiency of phage-based biocontrol [45,46,92,133,134], large virion morphogenesis yields are usually associated to lengthier latency periods [135].

The *in vitro* inactivation of planktonic Pcg IBSBF-158 cells by phage PcgS01F (as shown in Fig. 7a) was similar at all MOI assayed during the first 5 h of treatment, with nearly full control of bacterial growth, but after 6 h of incubation MOI 10, 100 and 1000 were the most successful in attaining expressive reductions in the bacterial load (Fig. 7a and Fig. 7b). After 12 h of treatment, using phage PcgS01F at MOI 1000 was highly effective in reducing the bacterial load by an astounding 82 % relative to the untreated control (Fig. 7a and Fig. 7b). Despite this, after 9 h of treatment the bacteria acquired mild resistance to the phage virions at all MOI assayed but, by the end of the treatment timeframe the bacterial load was still expressively reduced when compared to the untreated control (Fig. 7a).

Several studies have claimed that bacterial numbers are either reduced more with phage at increasing MOI or reduced quicker at a higher MOI [45–47,136–138]. In the study reported herein, increasing

MOI from 0.01 → 1 → 100 did significantly increase the efficiency of phage treatment (Fig. 7a and Fig. 7b), after a period of 12 h.

Regardless major concerns related to using phages in biocontrolling (phyto)pathogens due to the emergence of mutant cells resistant to the phage(s) with concomitant regrowth of bacteria following phage-treatment [35,96], several researchers around the globe argue that, because of the large variability in both the number and location of phage receptors on the surface of target bacterial cells, bacterial regrowth after treatment with phage virions can be circumvented by the use of a cocktail integrating different lytic phages with distinct adsorption/desorption mechanisms [35,36,92,93,139–141]. In the study entertained herein, for all MOI tested except MOI 1000, the bacteria started to regrow after 8 h of treatment, consistent with the frequency of emergence of phage-resistant bacterial mutants (Table 2), with significant regrowth after 12 h of treatment (Fig. 7a) for the lowest MOI.

Although the results obtained in the present work for the frequency of emergence of spontaneous bacterial mutants resistant to phage PcgS01F (viz. 6.194×10^{-4} , Table 2) are in close agreement with results published in the literature for the same phytopathogen in the presence of phage ph002F (6.793×10^{-4}) or phage ph004F (1.162×10^{-3}) [33] as well as for other *Pseudomonas* species, viz. *P. aeruginosa* strain PAO1 in the presence of phage PA5P2 (1.7×10^{-3}) [142], the spontaneous emergence of resistant mutants of Pcg cells in the presence of phage PcgS01F is 1–2 orders of magnitude higher than results published for *P. syringae* pv. *porri* in the presence of phage KIL3 (1.83×10^{-6}) or phage KIL4 (3.33×10^{-6}) [143], for *P. aeruginosa* strain PAO1 in the presence of phage PaP1 (3.0×10^{-5}) [144], and for *P. aeruginosa* strain K (PAK) in the presence of phage LUZ19v (1.00×10^{-7}) or phage PAK_P1 (6.00×10^{-5}) [145]. Upon exposure to phage virions, several authors have suggested a decay in bacterial fitness as a putative explanation for the faster removal of these out-of-shape cells from the environment when compared with their wild-type counterparts [146,147]. More recently, Siström et al. [148] demonstrated that *P. syringae* pv. *phaseolicola* cells become resistant to phage $\phi 6$ by eliminating its type IV pili that phage $\phi 6$ uses to attach onto, with a concomitant reduction in virulence of such mutant cells resistant to the phage.

The study of the impact of pH (5.0, 7.0, and 10.0) upon the lytic activity of phage PcgS01F allowed one to observe that the phage titre decreased at both acidic and basic pH values (as shown in Fig. 8a); however, there were no differences among pH values 5.0 and 10.0 up to 4 h of incubation, after which phage PcgS01F lytic activity decreased more at pH 10.0 than at pH 5.0 up to 12 h of incubation (Fig. 8a, ANOVA, $p < 0.05$). At pH 5.0, phage PcgS01F lost ca. 0.6 log PFU/mL, whereas at pH 10.0 it lost 0.5 log PFU/mL, over a 72 h period of incubation, differences that were not statistically significant (Fig. 8a, ANOVA, $p > 0.05$). At pH 7.0, phage PcgS01F lost a little amount of lytic activity during the first 2 h of incubation (viz. ca. 0.2 log PFU/mL) but after this initial period it endured until 12 h of incubation without any detectable loss of activity (Fig. 8a, ANOVA, $p > 0.05$), after which it lost only ca. 0.3 log PFU/mL until the end of the 72 h timeframe of assay. Hence, phage PcgS01F revealed to be quite resistant to the effects of pH upon its lytic activity. These results compare with results published by Pinheiro et al. [35] for phage phi6 (lytic against *Pseudomonas syringae* pv. *syringae*, Pss) exposed to pH 6.5, 7.0 and 7.5, where a maximum loss of 2 log PFU/mL over a period of 68 h was noticed. Compared to phages ph002F and ph004F (yet unpublished data) phage PcgS01F revealed a higher resistance to the effects of pH, with phage ph002F losing more than 2 log PFU/mL at pH 5 and 10, and phage ph004F losing ca. 2 log PFU/mL at pH 5 after 72 h and all activity at pH 10 after 36 h.

The effect of temperature upon phage PcgS01F lytic activity also revealed that this phage was very stable at 4 °C, 17 °C and 25 °C, without any loss of activity at these temperatures during the first 48 h of incubation (Fig. 8b, ANOVA, $p > 0.05$). After this timeframe, a minor decrease of lytic activity was noticed (viz. 0.4 log PFU/mL) at these temperatures up to 72 h of incubation (as shown in Fig. 8b). At 40 °C, the same trend of virion lytic stability was observed until the first 12 h of

incubation (Fig. 8b). After this time period, however, phage PcgS01F lost ca. 1 log PFU/mL of lytic activity. Hence, after 24 h of incubation, the reduction in the concentration of viable phage virions was much higher at 40 °C than at 4 °C, 17 °C or 25 °C (Fig. 8b, ANOVA, $p < 0.05$). These results compare with results published by Pinheiro et al. [35] for phage phi6 exposed to temperatures 15 °C and 25 °C, where a maximum loss of 1.5 log PFU/mL over a period of 68 h was noticed whereas at 37 °C it lost all of its activity after only 6 h of exposure. Compared to phages ph002F and ph004F (yet unpublished data) phage PcgS01F revealed a higher resistance to the effects of temperature, with phage ph002F losing more than 1.5 log PFU/mL at 41 °C, and phage ph004F losing ca. 3.5 log PFU/mL at 41 °C after 72 h. These two phages were completely stable at 4, 17 and 25 °C (data not yet published).

Solar radiation (specifically, its UV component) is most likely the leading factor that promotes loss of phage infectivity in the environment [35,55], with short (UV-B) radiation wavelengths damaging irreversibly the phage virion genome and altering viral proteins with concomitant production of (lethal) photoproducts for the virions. When phage PcgS01F was exposed to solar radiation, the abundance of phage lytic activity endured throughout the exposure timeframe, with only a marginal difference between control and test (Fig. 8c, ANOVA, $p > 0.05$). Such difference was not statistically significant, meaning that there is not sufficiently strong evidence to reject the null hypothesis that there is no difference between test and control groups, and so one cannot conclude that the groups are different. The titre of lytic virions (SR) did not change (Fig. 8c) after 7 h of exposure, when compared with the phage control (SR-C). Phage PcgS01F revealed, therefore, to be quite resistant to direct sunlight exposure. These results compare with results published by Pinheiro and colleagues [35] for phage phi6 exposed to solar radiation during 6 h, where a maximum loss of 2 log PFU/mL was noticed. Compared to phages ph002F and ph004F (yet unpublished data) phage PcgS01F revealed a higher resistance to the effects of ultraviolet radiation, with phage ph002F losing ca. 2 log PFU/mL after a 6 h exposure, and phage ph004F losing nearly 3 log PFU/mL after a 4 h exposure and all activity after 5 h of exposure to solar radiation.

Further characterization by whole genome sequencing confirmed that phage PcgS01F is a siphovirus (phage morphotype with a long, flexible, non-contractile tail) (Figs. 3d–3e, Table 3). Phage PcgS01F belongs to the class Caudoviricetes (Figs. 3d–3e), family Drexlerviridae, genus *Guelphvirus* (Table 3), according to recent ICTV bacterial viruses' taxonomy [149]. The taxonomic lineage of phage PcgS01F was thus established as *Viruses* > *Duplodnaviria* > *Heunggongvirae* > *Uroviricota* > *Caudoviricetes* > *Drexlerviridae* > *Braunvirinae* > *Guelphvirus* [150]. The genome of phage PcgS01F was screened for the presence of integrase genes [151] and did not feature any putative lysogenic-related (integrase) sequences; since more than half of the predicted protein-coding genes have unknown function, one cannot conclude definitely that phage PcgS01F has a strictly lytic lifestyle [48,125]. The coding sequences (CDS) with predicted functions among the 76 annotated CDS in the genome of phage PcgS01F (Table 3) did not encode depolymerase enzymes, toxins, virulence factors or antibiotic resistance (Supplementary Table S1; Supplementary Table S2) (Fig. 9).

Siphoviruses of the genus *Guelphvirus*, family Drexlerviridae, are characterized by several unique features, viz. non-enveloped, head-tail structure, with an icosahedral head about 60 nm in diameter, long non-contractile and very flexible tail (ca. 151 nm length and 8 nm width), with rosette-like structures at the tip encompassing 4 short kinked terminal fibers which are involved in host recognition and attachment [152], with linear dsDNA genome of ca. 40–50 kb containing ca. 80 genes [153,154]. *Guelphvirus* is a genus of lytic bacteriophages that infect *Escherichia coli* and other Enterobacteriaceae [152], belonging to the class Caudoviricetes and family Drexlerviridae [152], a family of bacteriophages that infect Gram-negative bacteria, mainly Enterobacteriaceae and Pseudomonas. These structural and genomic features (Table 3) are well observed in the TEM photomicrographs in Fig. 3d,e.

The basic mechanical properties of genomic DNA play a pivotal role

in governing various aspects of its maintenance within cells, its functionality, and its interactions with other molecules, especially proteins. The degree of mechanical flexibility of the DNA molecule is determined by different factors such as the constraints imposed by the DNA backbone, the constituting sequence bases, and the interactions involved in their stacking [155].

DNA bendability can be analysed across different scales, ranging from the properties of individual base-pair steps to the broader behaviour of the double helix. This bendability is a characteristic that impacts the packing of genomic material and the regulation of its transcription, and can be exemplified in viral genomes, where significant bending occurs to accommodate the viral DNA within the capsid [156]. During such processes, transcription factors induce bending in the DNA sequence, forming complexes with proteins which in turn govern gene expression [157] with concomitant protein production. Increasing evidence suggests that variations in DNA sequence can influence its flexibility by altering the stacking interactions between neighbouring base pairs [158]. One approach to quantifying DNA bendability or flexibility concerning its biological function is by assessing the efficiency of dsDNA looping or cyclization.

The DNA cyclization depends on the length of the DNA molecule, a concept that was first investigated by Shore and Baldwin [159], demonstrating the helical structure of DNA through the periodic nature of cyclization efficiency [160]. More recently, Basu et al. [84] introduced a high-throughput technique named loop-seq to measure the intrinsic rate of DNA cyclizability. Their approach involved determining the cyclizability by calculating the logarithm of the ratio between the probability of encountering sequences within the looped group versus a control group, defining intrinsic cyclizability as the average value obtained by repeating the loop-seq procedure multiple times.

In our study, we explored the intrinsic cyclizability of the PcgS01F phage genome with a resolution down to 7-base pairs, following the methodologies outlined by Basu et al. [82]. Their method involved assessing intrinsic cyclizability across a wide range of 90,000 unique 50-base-pair DNA sequences. In the PcgS01F phage genome, certain peaks reach a maximum value of 0.712. This observation suggests potential differences in cyclizability within specific regions of the PcgS01F phage genome. On the other hand, the lowest peak in the PcgS01F phage genome is at -0.785 (Fig. 15a), indicating the possibility of more regions with reduced cyclizability in the genome. However, the mean cyclizability value for the PcgS01F phage genome is -0.14319 , a value that lies between those reported by Silva et al. [43] for two myoviruses (with an average of -0.12567 for the PsgM02F phage genome and -0.16482 for the PsgM04F phage genome).

It is essential to acknowledge that DNA looping primarily occurs in short-base pair sequences, due to their inherent capability to loop. Our findings revealed the local intrinsic cyclizability values within the phage PcgS01F genome, highlighting particular regions of the genomes with either high or low cyclizability (Fig. 15c). While this analysis effectively describes cyclizability at a local level, it doesn't serve as a global mechanical property of the entire phage genome.

Based on this observation, we conclude that the PcgS01F phage genome exhibits specific regions with higher cyclizability than the PsgM04F phage genome [43] but lower than the PsgM02F phage genome [43]. However, the majority of regions within the PcgS01F phage genome demonstrate an average intrinsic cyclizability around -0.14319 , compared to the -0.12567 cyclizability for the PsgM02F phage genome and -0.16482 cyclizability for the PsgM04F phage genome [43]. Consequently, this suggests that, on average, the PcgS01F phage genome possesses more flexible regions than the PsgM04F phage genome but less flexible regions than the PsgM02F phage genome. These flexible regions are likely influenced by the DNA shape discussed in the section above, such as ProT, MGW, HelT, and Roll. Altogether, these results are in clear agreement with previous findings from our research group, with a more flexible phage genome leading to a higher virion morphogenesis yield. In this study, phage PcgS01F produced 46

virions/host cell (Fig. 6) whereas phage PsgM02F produced 124 virions/host cell and phage PsgM04F produced 12 virions/host cell [43].

Several experiments have indicated a connection between DNA flexibility or cyclizability and specific dinucleotide steps at the local level. A dinucleotide refers to the pairing of two nucleotides along the DNA sequence. The presence of particular dinucleotides within a sequence can be correlated with the geometric and structural characteristics of the DNA [161]. The frequency of occurrence of a dinucleotide in a sequence is commonly utilized as a genomic signature for various microorganisms [162].

Various studies have established the influence of dinucleotides and their spatial configuration within the genome in shaping a more flexible DNA structure [155,161,163,164]. Basu et al. [82] studied how this spatial organization of specific dinucleotides can impact intrinsic cyclizability at the local level, by employing high-throughput data to compute the pairwise distance distribution function which evaluated the frequency of distance correlation for each dinucleotide step and its association with DNA mechanics, such as cyclizability, over 100 steps. In their study, Basu et al. [82] explored the impact of 136 dinucleotide steps (NN-NN) on the intrinsic DNA cyclizability at the local level. These researchers discovered that dinucleotide pairs composed entirely of adenines or thymines positively contributed to intrinsic cyclizability and, at the same time, a similar effect was observed for dinucleotide pairs consisting only of guanines or cytosines.

Basu et al. [82] highlighted that those dinucleotides composed entirely of A's, T's, G's, and C's exhibit a positive contribution to intrinsic cyclizability at the local level. Upon closer examination of these dinucleotides in our findings, we observed that the occurrence of the AA and TT dinucleotides is higher in the PcgS01F phage genome compared to those of AC, AG, CC, CG, GA, GG and GT (Fig. 14). We opted to investigate the frequency of occurrence of each dinucleotide along the phage genomes without considering distance or arrangement within the genome. Basu et al. [82] reported that the TA dinucleotide had the most significant positive influence, while CG was identified as the least flexible dinucleotide concerning intrinsic cyclizability. However, TA emerges as a dinucleotide with a relatively high occurrence in the PcgS01F phage genome (Fig. 14). While Packer et al. [121] have previously established that TA can impart special mechanical properties like DNA flexibility, recent studies such as those by Back and Walther [165] further support this observation. Back and Walther [165] found a correlation between the TA dinucleotide and DNA cyclizability, giving significance to its strong association with DNA flexibility. These studies might suggest that an increased presence of TA's tends to make the DNA more bendable on average, whereas a higher proportion of CG's tends to render the DNA less bendable on average. In the results obtained (Fig. 14), CG appears with a relatively low frequency, whereas TA appears with much higher frequency, hence hinting to a degree of flexibility of the PcgS01F phage genome.

Our findings revealed a TA occurrence frequency of 5.71 % and a CG frequency of 4.09 % (Fig. 14). This observation suggests the possibility of a certain flexibility for the PcgS01F phage genome compared to the PsgM04F phage genome [43] in regions where this dinucleotide is prevalent. On the other hand, the CG dinucleotide may indicate a level of rigidity in the PsgM04F phage genome, particularly in regions where the CG dinucleotide is predominant [43]. In addition, Rao et al. [166] identified that cytosine methylation most frequently happens in CG dinucleotides, affecting the structure of the DNA sequence and local DNA shape, mostly Roll and ProT. These researchers found a strong methylation effect of 50–100 % of the range observed across all unmethylated sequences, increasing the Roll angle and decreasing the ProT angle in methylated CpG steps. Moreover, Basu et al. [82] reported that on a single DNA sequence, CpG methylation makes the DNA more rigid. Our results showed that CG dinucleotide has a not so low frequency among dinucleotides (Fig. 14), and therefore there is a possibility of methylation in this step in the PcgS01F phage genome, but at this moment this is a speculation that needs to be more studied.

Finally, we have to emphasize mentioning that our analyses (DNA shape and intrinsic cyclizability) are properties that greatly influence the DNA at a local or nucleotide level, but are not so determining at a global level, providing however, a great characterization of the PcgS01F phage genome at the nucleotide level and giving insight into the flexibility/cyclizability of this phage genome. We are aware that a global study of the phage genome would give a broader understanding, and we will take this into account for future studies.

In the *ex vivo* bacterial inactivation assays, phage PcgS01F infected and successfully killed Pcg cells in coffee plant leaves artificially contaminated. The difference in Pcg counts were statistically significantly different at all treatment times for both MOI 10 and MOI 1000 when compared to BC (Fig. 16a). Throughout the *ex vivo* experiments, when phage PcgS01F was incubated in the presence of its host, a statistically significant increase (3.94 log PFU/mL) in phage concentration was observed for both MOI 10 and 100 (Fig. 16b). Differently from the conditions prevailing in suspension, where phage virions and planktonic cells contact with each other via random encounters leading to productive cell infections, on the surface of coffee leaves the conditions are quite different with their (solid) waxy layer imparting limited degrees of freedom to both Pcg cells and phage virions, with concomitant reduced encounters and leading to much lower net phage progeny production. This is most likely the explanation for the lower increase in phage virions during the *ex-vivo* treatments.

In a previous work [33], two new myoviruses lytic for the same phytopathogen were quite effective at MOI 10 (reduction of 4.59 log CFU/mL by phage ph002F and reduction of 3.85 log CFU/mL by phage ph004F) after 10 h of incubation with the host. As a cocktail, the bacterial reduction promoted was even larger (5.26 log CFU/mL at MOI 0.1 or 5.03 log CFU/mL at MOI 10), after a 12h-treatment. In *ex vivo* phage-bacteria inactivation experiments, the maximum of Pcg inactivation was 5.36 log CFU/mL at MOI 100 by the cocktail of both phages after 6 h of incubation, and 5.11 log CFU/mL for MOI 100 after a 24h-treatment [33]. But, with those two myoviruses, a slight bacterial regrowth was observed until the end of the incubation timeframe, which did not happen with the newly isolated phage described herein at a higher MOI. Hence, a cocktail integrating the two myoviruses already characterized and the siphovirus described in the present work would be most likely highly effective in controlling the phytopathogen. This will be the subject of a forthcoming publication.

The reduction of Pcg concentration by phage PcgS01F in *in vitro* and *ex vivo* phage-bacteria inactivation experiments is an important advance towards development of an efficient alternative to the conventional copper/antibiotic-based treatments for bacterial halo blight of coffee plants. However, in order to translate the application of this strategy to real coffee plantations, *in planta* studies (both in greenhouse conditions and in the field) are needed using both whole coffee plant seedlings and adult coffee plants, together with a phage cocktail integrating several (different) lytic phages able to infect a wide range of virulent Pcg strains.

5. Conclusions

The results of the research work presented herein clearly suggest that treatment of infections in coffee plants by the phytopathogen *Pseudomonas coronafaciens* pv. *garcae* using phage PcgS01F has the potential to be an effective alternative to the conventional chemical control of that phytopathogen. However, selection of other lytic phages to integrate a lytic cocktail with a broad-host-range should allow to circumvent the propensity to promote the development of bacterial-acquired phage-resistance and prevent bacterial regrowth. The present work describes in detail a robust pipeline for the full characterization of a novel isolated phage targeting this type of application. The *in vitro* and *ex vivo* reduction of Pcg cell concentration by the new phage isolated and duly characterized is, nevertheless, an important leap forward in the development of an ecofriendly and efficient treatment for coffee plant bacterial blight.

Author agreement

I, the Corresponding Author, declare that this manuscript contains original, unpublished results and confirm that it has been read and approved by all the authors. I understand that the Corresponding Author is the sole contact for the Editorial process and is responsible for communicating with the other authors about progress, submissions of revisions and final approval of proofs.

Institutional review board statement

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CRediT authorship contribution statement

Basilio Cieza: Writing – review & editing, Software, Methodology, Formal analysis, Conceptualization. **Aakash Basu:** Software, Methodology, Formal analysis, Conceptualization. **Lucas M. R. Rodrigues:** Validation, Methodology, Formal analysis. **Marta M. D. C. Vila:** Writing – review & editing, Formal analysis. **Victor M. Balção:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Resources, Project administration, Methodology, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Erica C. Silva:** Writing – original draft, Methodology, Investigation. **Carlos A. Quinde:** Validation, Software, Methodology, Formal analysis, Conceptualization. **Luan C. Mota:** Writing – original draft, Methodology, Investigation.

Declaration of Competing Interest

The authors declare no conflict of interest. The funders had no role in the design of the study; in the collection, analyses, or interpretation of data; in the writing of the manuscript; or in the decision to publish the results.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the

online version at [doi:10.1016/j.enzmitec.2024.110573](https://doi.org/10.1016/j.enzmitec.2024.110573).

Data availability

The phage genome sequence described in this work has been deposited in GenBank NCBI (National Center for Biotechnology Information) under accession number PP134917 (phage PcgS01F)..

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